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Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed.
We post it as supplied by the authors.

Supplement to: Moss-Morris R, Norton C, Hart A, et al. Digital cognitive behavioural self-management programme for fatigue, pain, and faecal incontinence in inflammatory bowel disease (IBD-BOOST): a multicentre, parallel, randomised controlled trial. *Lancet Digit Health* 2025. https://doi.org/10.1016/j.landig.2025.100906

Online supplementary material for:

Digital cognitive behavioural self-management programme for fatigue, pain and faecal incontinence in inflammatory bowel disease: the IBD-BOOST randomised controlled trial

Contents **Pages**

A Randomised Controlled Trial of supported online self-management for symptoms of fatigue, pain and urgency/incontinence in people with inflammatory bowel disease: the IBD-BOOST trial

Version 6.0 Date: 21.03.2022

IRAS number: 258725

Project Sponsor: London North West University Healthcare NHS Trust
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PROJECT INVESTIGATORS

Chief Investigator: Professor Christine Norton, King's College London (KCL)

Co-investigators: [see list on page 6]

The study as detailed within this research protocol **(Version 6.0, 21.03.2022)**, or any subsequent amendments will be conducted in accordance with the UK Policy Framework for Health and Social Care Research 2017, the World Medical Association Declaration of Helsinki (1996) and the current applicable regulatory requirements and any subsequent amendments of the appropriate regulations.

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PARTICIPATING SITES - SIGNATURES

The principle investigator at each site is required to sign a separate signature (Appendix 2), confirming that they have received, read and understood the protocol requirements as set out in this document. A copy of the fully-signed protocol with these signatures will be stored in the Investigator Site File at each site, and in the Trial Master File.

ABBREVIATIONS / GLOSSARY OF TERMS

AE	Adverse Event
CCUK	Crohn’s & Colitis UK (registered charity supporting people with IBD)
CD	Crohn’s disease
CI	Chief Investigator
FI	Faecal Incontinence
GDPR	General Data Protection Regulation
HRA	Health Research Authority
IBD	Inflammatory Bowel Disease
IBS	Irritable Bowel Syndrome
ISF	Investigator Site File
KCL	King’s College London
LNWH	London North West Healthcare NHS Trust
NHS	National Health Service
NICE	National Institute for Clinical Excellence
NIHR	National Institute for Health Research
NRES	National Research Ethics Service
PCTU	Pragmatic Clinical Trials Unit (QMUL)
PGfAR	Programme Grant for Applied Research
PI	Principal Investigator (local collaborator / lead at each research site)
PIL	Patient Information Leaflet
PMG	Programme Management Group
PPI	Patient and Public Involvement
PSC	Programme Steering Committee
PPI	Public and Patient Involvement
QMUL	Queen Mary University of London
RCT	Randomised controlled trial
REC	Research Ethics Committee
SAE	Serious Adverse Event
SWAT	Study within a trial
TMF	Trial Master File
UC	Ulcerative Colitis

STUDY SUMMARY

<table>
<tr><td>Short Title</td><td>The IBD-BOOST Trial</td></tr>
<tr><td>Methodology</td><td>Randomised controlled trial with embedded health economics evaluation and process evaluation and a recruitment “SWAT” (study within a trial)</td></tr>
<tr><td>Research Sites</td><td>NHS Trusts (subject to site feasibility assessments and continued capacity):
1. London North West University Hospital NHS Trust
2. Nottingham University Hospitals NHS Trust
3. The Queen Elizabeth Hospital King’s Lynn NHS Foundation Trust
4. St Helens and Knowsley Teaching Hospitals NHS Trust</td></tr>
<tr><td>Objectives/Aims</td><td>To answer the primary research question: In people who report symptoms (scoring at least 5/10 on one or more symptoms on a 0-10 scale) and express a desire for intervention, does a facilitator-supported tailored (to patient needs) online self-management programme for fatigue, pain and faecal urgency/incontinence in IBD improve IBD-related quality of life and global rating of symptom relief 6 months after randomisation, compared with care as usual?</td></tr>
<tr><td>Number of Participants/Patients</td><td>At least 740 participants</td></tr>
<tr><td>Main Inclusion Criteria</td><td>People wanting interventions for symptoms of fatigue, pain and/or urgency, recruited from people responding to our earlier IBD-BOOST survey (Stage 2 of the programme) or from people who have completed the survey plus medical symptom optimisation study (IBD-BOOST Optimise - Stage 3 of the programme), who request further intervention for these symptoms and the impact of at least one of these 3 symptoms was scored 5 or more on a 0-10 scale.

Patients with IBD who meet the following:
Inclusion criteria:
• Diagnosis of IBD (self-reported as having been medically diagnosed with IBD including patients with an ileo-anal pouch or stoma)
• 18 years old or over
• Living in England, Scotland or Wales
• Have participated in Stage 2 of the programme (IBD-BOOST survey) and have self-scored the impact on quality of life of one or more symptoms of fatigue, pain or urgency/incontinence as 5 or more on a 0-10 scale when completing Stage 2 (IBD-BOOST survey) or Stage 3 (IBD Optimise) (whichever is the more recent)
• No “red flags”– see below
• Access to the online intervention via a computer or mobile device

Exclusion criteria:
• One or more “red flags” identified on pre-randomisation screening, (such as new bleeding, rapid weight loss or vomiting) self-reported on a screening checklist (Appendix 4)
• Inability to give informed consent (for example, due to reduced mental capacity)
• Insufficient command of English to understand study documents and procedures
• No access to online materials</td></tr>
<tr><td>Methodology and Analysis (if applicable)</td><td>Two-arm randomised controlled superiority trial with an internal pilot study, a study within a trial (recruitment and retention questions), economic evaluation, and a process evaluation.
Primary analysis: Comparison of the primary outcomes (quality of life (UK-IBDQ) and global rating of satisfaction scale) between the intervention and usual care arms at 6 months after randomisation on intention-to-treat basis. Analysis models are adjusted for baseline values of the outcome measure, stratification variables and clustering by facilitator in the intervention arm</td></tr>
<tr><td>Proposed Start Date</td><td>1st December 2019</td></tr>
<tr><td>Proposed End Date</td><td>31st July 2022</td></tr>
</table>

Proposed follow-up end	30th June 2023
Study Duration	42

LIST OF CONTENTS

1. Background

Inflammatory Bowel Disease (IBD) affects 300,000 people in the UK (Crohn's & Colitis UK, 2016), causing unpredictable bouts of gut inflammation, with acute illness, diarrhoea, and pain. In remission, many people with IBD live with fatigue, chronic abdominal pain, and bowel urgency/incontinence (1). There is no current cure for IBD, which usually starts in childhood or as a young adult. Most previous IBD research has focused on controlling inflammation. However, many people report continuing IBD-related fatigue (41%), abdominal pain (62%) and difficulty with continence (up to 75%) even when IBD is in remission (1-3). These symptoms limit peoples' quality of life and ability to work and socialise. Patients feel that these symptoms are not taken seriously by health professionals and report that little help is given (2, 4, 5). However, the James Lind Alliance IBD research priority-setting consensus put fatigue, pain, and incontinence in the top 10 issues that IBD patients and clinicians want to be addressed by research (6).

1.1 The IBD-BOOST programme of research

The current work package is stage four of IBD-BOOST, a National Institute for Health Research (NIHR) Programme Grant for Applied Research (PGfAR) funded programme. The overall aim of the Programme Grant is to improve the quality of life of people with IBD by reducing the burden of IBD-related fatigue, abdominal pain, and urgency/incontinence.

This protocol is for the final part of the programme only, a randomised controlled trial (RCT) of an online intervention for people who have expressed a desire for intervention for fatigue, pain and/or urgency/incontinence.

Stage 1 of the programme involved focus groups and interviews with people with IBD and IBD nurse specialists to inform the development of the intervention under investigation in this proposed trial. This study was undertaken with REC Approval (REC reference: 17/WA/0349/ IRAS number: 228902). We have subsequently used this data to inform development of the online self-management intervention that will be tested in this RCT and are feasibility testing (March-May 2019).

Stage 2 of the programme involves a large cross-sectional survey to people with IBD to investigate the inter-relationships of these IBD-related symptoms and the proportions wanting support to manage these symptoms. This survey protocol has been submitted for REC approval (REC reference: 18/NW/0613/ IRAS number: 246185). The protocol for this survey states: *Participants who consent to be contacted for further research and are eligible will be invited to participate in the RCT* (stage 4).

Stage 3 of the programme is a non-randomised experimental study to test the effectiveness of a checklist and algorithm for identifying and treating medical causes of these IBD-related symptoms. We will then treat any medical abnormalities detected. Participants who complete this and have continuing symptoms after medical optimisation and are eligible (see RCT eligibility criteria below) will be invited to take part in this trial. This study has been submitted to a REC for ethics approval (IRAS number 249845;19/WM/0107).

Stage 4 (the current proposal) is an RCT of online self-management for these symptoms of IBD fatigue, pain and urgency/incontinence, with an embedded pilot study, a study within a trial, health economics evaluation and process evaluation. Potential participants will already have completed the IBD-BOOST survey (Stage 2). Some of them will also have participated in Stage 3. See Appendix 1 for overall plan for the IBD-BOOST programme of studies.

1.2 Rationale

These symptoms of fatigue, pain and urgency/incontinence have a major impact on quality of life in people with IBD but have been largely ignored by clinicians and researchers. Our programme, shaped by the concerns of our patient and clinician stakeholders, focuses on a supported online self-management intervention for these symptoms.

1.3 The need for research

There has been remarkably little research on managing these troublesome IBD symptoms. Cognitive behavioural (CB) theories of these symptoms in other conditions suggest that disease factors trigger symptoms, but an interaction of cognitive, behavioural, emotional and physiological responses may strengthen and perpetuate them (7-10). For instance, believing that pain and fatigue have serious consequences (cognitions), leads to avoidance of activity (behaviour) and distress (emotion). Distress activates the autonomic nervous system (physiology) leading to poor sleep and symptom magnification. CB-based interventions have been shown to reduce symptom severity and improve quality of life in other diseases (9, 11, 12). Online delivery of these interventions for chronic pain and fatigue appear effective, with evidence of enhanced effects through the addition of minimal health care professional support (13-17).

Empirical modelling work and systematic reviews (in preparation for this study) of risk factors and interventions for IBD fatigue (2, 18) and of interventions for chronic abdominal pain in IBD (19) suggest that similar interventions, encouraging patients to alter emotional, behavioural and/or cognitive responses to these symptoms, will be effective (20). According to the Common Sense Model (21), patients generate cognitive and emotional representations (illness perceptions) in response to their illness. Illness perceptions provide a framework for IBD patients to make sense of their illness and in turn guide behavioural coping strategies, e.g. resting or avoidance of activities, with potential impact on outcomes (22-24). In line with this model, negative illness perceptions and related avoidance behaviours are associated with quality of life in IBD (25, 26). Our recent empirical modelling work showed that negative illness perceptions (including believing fatigue is uncontrollable and has very negative consequences), and coping with fatigue through daytime napping and all-or-nothing avoidance behaviours explained 41% of variance in IBD-related fatigue after controlling for disease status (18). There have been no high quality treatment trials for fatigue in IBD (27). All-or-nothing behaviour is a pattern of behaviour where people do too much when they feel less fatigued, feel extreme fatigue as a result and avoid activity until fatigue reduces (18). We have been funded by Crohn's & Colitis UK (CCUK) to test a similar CB model of abdominal pain in IBD. Our previous work on fatigue and pain in Multiple Sclerosis (10, 28) suggests that illness and symptom cognitions and avoidance behaviours are likely to be related to pain in IBD.

For incontinence, pelvic floor and bowel retraining are effective in people without IBD (29, 30) irrespective of intervention modality (31, 32). Diet and anti-diarrhoeal medication also help (31, 33, 34). People with urgency enter a vicious circle of hypersensitivity to rectal sensation, constant monitoring of rectal contents, and anxiety in anticipation of incontinence (35). Anxiety stimulates gut peristalsis and triggers a rush to the toilet; running makes incontinence more likely. Breaking that circle by bowel retraining (practicing "holding on") re-builds confidence and ability to defer defecation (36-38). Of people with IBD-related incontinence, 58% feel that anxiety worsens incontinence; even those who never experience incontinence worry about the possibility (3, 5). CB methods based on a similar vicious cycle of symptom focusing and resultant anxiety improves symptom severity and impact in patients with irritable bowel syndrome (IBS) (39). Simple instructions and a self-help booklet works as well as complex biofeedback for improvement in a non-IBD population (29). The results of our current trial of a self-management paper booklet vs. booklet plus facilitator support for incontinence in IBD (40) has informed the intervention development in this programme. Providing education and training in these behavioural methods could be embedded within a broader CB intervention and has not previously been tried in IBD (41).

People with IBD have been found to engage with, and complete, online self-management interventions (42). In a 12-month trial in 333 patients, 79.6% completed the study, 88% finding this feasible and preferable to face-to-face care (43). However, education alone decreases hospital attendance and time to treat relapse, but does not improve quality of life (44, 45), even if better knowledge is associated with reduced costs (46). People with long term conditions, including IBD, need health professional support to help them engage with self-management (47). People with IBD want to take a greater role in self-management, including information that is both therapeutic and supportive (48). Previous studies of online CBT in IBD have shown significantly improved quality of life, even when participants were recruited with no specific symptoms (49). Two ongoing trials are testing CBT for mental health issues in IBD (50).

In other chronic diseases, self-management improves cognitive symptom management and reduces fatigue, distress and social limitations (51). CB models, theory and principles are ideally suited to self-management interventions, providing a clear structure to develop theoretical models (including specific cognitions and behaviours) related to the outcome of interest (e.g. quality of life related interference of symptoms), and specific techniques to enhance behavioural and cognitive change. Many techniques are available, including self-assessment of behavioural patterns, setting and monitoring of goals (52). Others are specific to CBT such guided discovery (enabling people to identify strengths and problem areas) and identifying and challenging unhelpful thought patterns (including negative illness beliefs) (53).

This trial will compare an online delivered self-management programme using the principles of CBT with telephone and online messaging support to standard care. The trial will provide evidence for the effectiveness of online delivered self-management programme to improve quality of life for those with IBD-related symptoms of fatigue, pain and urgency, enabling clinicians and patients to make an informed decision regarding management.

The Medical Research Council (MRC) has funded a programme of work named "PROMETHEUS" promoting the use of "Studies within a Trial" (SWATs) to answer some important methodological issues about recruitment and retention in RCTs (https://www.nihr.ac.uk/funding-and-support/funding-for-research-studies/studies-within-a-trial.htm). We have been invited to participate and will also attempt to answer a methodological question in our RCT. The embedded SWAT ('PROMETHEUS in IBD BOOST') aims to evaluate the impact on participant recruitment of a brief participant information leaflet (PIL) compared with a standard length PIL. This will be implemented using a RCT design. Each patient being invited into the IBD BOOST RCT will be randomised (separate randomisation before the main RCT randomisation) to one of the following: 1) the standard length IBD BOOST RCT PIL; 2) a brief PIL, with links for more detailed information for participants to access. We will evaluate whether receiving the brief PIL is associated with higher levels of recruitment into IBD BOOST. The full protocol for PROMETHEUS in IBD BOOST is attached as Appendix 3.

2. Research plan

This application covers a randomised controlled trial including the following:

- Trial internal pilot study
- Randomised controlled trial
- Study within a trial
- Heath economics evaluation
- Process evaluation

2.1. Internal pilot study

Aim

To test planned methods and procedures for the main RCT and identify barriers to the components working together and potential recruitment issues.

Objectives:

- Train to competence at least 5 facilitators
- Recruit and randomise the first 100 participants for the RCT, aiming to complete this within the first 12 months of the RCT recruitment
- To identify likely recruitment rates and barriers to recruitment
- To test recruitment and retention assumptions
- To test the feasibility of completing the IBD Resource Use Questionnaire (incorporated into the main study outcome measure Appendix 11) online (by assessing data quality and completion rate).

The methods planned for the main RCT below will be followed. If no substantive changes are made these 100 participants will be included in the main RCT analysis.

The programme management group and the steering group will review achievement of the objectives above (not outcomes of the intervention) to decide if any adjustments are needed to study processes. This will be decided at a minuted meeting (possibly a specially convened teleconference or online meeting) of the Steering Group who will decide on any changes proposed by the programme management group.

2.2: Randomised controlled trial (see Figure 1: consort diagram and Table 1: Screening and data collection schedule)

2.2.1 Primary research question:

1. In people who report symptoms (scoring at least 5/10 on one or more symptoms on a 0-10 scale) and express a desire for intervention, does a facilitator-supported tailored (to patient needs) online self-management programme for fatigue, pain and faecal urgency/incontinence in IBD improve IBD-related quality of life and global rating of symptom relief 6 months after randomisation, compared with care as usual?

2.2.2 Secondary research questions:

2. Is there any difference between the groups in severity of symptoms of fatigue, pain and urgency/incontinence at 6 and 12 months after randomisation?
3. Is there any difference between the groups in IBD-related quality of life and global rating of symptom relief 12 months after randomisation?
4. Does prior medical optimisation of symptoms (in Stage 3 of this programme) moderate the treatment response (i.e.do those receiving medical optimisation show greater treatment gains than those who don't) as measured by the primary outcomes?
5. Do people with quiescent IBD at trial commencement have a better response to treatment (primary outcomes) than those with active disease? (quiescent disease defined as faecal calprotectin within normal range of 200µg/g or less and/or IBD-control score of 13 or more)?
6. Does baseline depression, or the presence of Irritable Bowel Syndrome (ROME IV criteria) moderate treatment response to intervention (primary outcome measures)?
7. Do changes in illness perceptions and behaviours, IBD specific anxiety and self-efficacy, and depression from baseline to 6 months mediate intervention effects on the primary outcomes at 12 months?
8. Is a facilitator-supported tailored online self-management programme for fatigue, pain and faecal urgency/incontinence in IBD cost-effective?
9. What are patient's expectations and experiences of the intervention and what factors may have influenced the intervention implementation (explored in process evaluation)?

2.2.3 **Hypothesis for Randomised Controlled Trial**

A facilitator-supported, online, tailored self-management programme for fatigue, pain and faecal urgency/ incontinence in people with IBD will result in better quality of life compared to care as usual at 6 months after randomisation.

2.2.4 Study Design

A pragmatic multi-centre two-arm, parallel group superiority RCT, with an internal pilot, of facilitator-supported online self-management versus care as usual (CAU) to manage symptoms of fatigue, pain, and faecal urgency/incontinence in IBD (See Figure 1; IBD-BOOST Trial consort diagram page 18). One baseline assessment and two assessments at 6 and 12 months after randomisation. Primary outcomes: IBD quality of life (UK-IBDQ scale) and global rating of symptom relief scale at 6 months as multiple primary outcomes.

2.2.5 Participants

People with symptoms of fatigue, pain and/or urgency, recruited from people responding to our earlier IBD-BOOST survey (Stage 2 of the programme) or from people who have completed the survey plus medical symptom optimisation study (Stage 3 of the programme), who request further intervention for these symptoms and who have scored the impact of at least one of these 3 symptoms as 5 or more on a 0-10 scale.

Patients with IBD who meet the following:

2.2.6 Inclusion criteria:

- Diagnosis of IBD (self-reported as having been medically diagnosed with IBD including patients with an ileo-anal pouch or stoma)

- 18 years old or over
- Living in England, Scotland or Wales
- Have participated in Stage 2 of the programme (IBD-BOOST survey) and have self-scored one or more symptoms of fatigue, pain or urgency/incontinence as having an impact on their life of 5 or more on a 0-10 scale when completing Stage 2 (IBD-BOOST survey) or Stage 3 (medical symptom optimisation) (whichever is the more recent)
- No “red flags”– see below
- Access to the online intervention via a computer or mobile device

2.2.7 Exclusion criteria:

- One or more “red flags” identified on pre-randomisation screening, (such as new bleeding, rapid weight loss or vomiting that has not been previously reported to a health care practitioner) self-reported on a screening checklist (Appendix 4). Note: The “red flags” criteria have been developed in consultation with 5 consultant gastroenterologists who are either co-applicants or part of our wider advisory group: they have each confirmed that they feel that participants who meet the criteria will be safe to enter an online self-management programme. If ineligible because of a red flag, a participant may be re-assessed if they contact the research team and report the information they have provided has changed such as the symptom has been adequately investigated or managed, in which case the participant can be included (see below).
- Inability to give informed consent (for example, due to reduced mental capacity)
- Insufficient command of English to understand study documents and procedures

2.2.8 Screening for eligibility

The central IBD-BOOST research team will screen the IBD Survey (stage 2) responses and the follow up questionnaire of participants in the IBD-BOOST Optimise study (stage 3) who have consented to be contacted for further research. Using the eligibility criteria above, potential participants will be identified. Sufficient English and capacity for consent will be presumed from the previous response to the survey. They will then be randomised to receive the standard or short PIL (see SWAT below). This will be embedded in an email from a member of the central research team, as will a link to the online consent form for those who wish to participate in the RCT.

If the online database is not available, paper copies of the standard PIL will be sent with a letter and a consent form. Participants receiving paper copies will be not be included in the SWAT.

2.2.9 Consenting

Written information about the RCT (Appendix 5 for full PIL, Appendix 6 for brief version: see SWAT below) and an invite email/letter (Appendix 7) will be emailed or posted to potential participants who have previously completed the IBD-BOOST Survey (stage 2 of the research programme) and have consented to further approaches to participate in research. Participants will be able to participate by providing consent online on the study website (Appendix 8) or by returning a completed consent form by post. See Section 2.3 below for information on the Study within a trial (SWAT) which will be testing a standard versus brief participant information leaflet during recruitment to this RCT. There will be two reminders by text or email (Appendix 9) after 10 and 20 days for non-responders.

All participants are free to withdraw from the study at any time without giving reasons and without prejudicing further treatment. Their right and access to their usual NHS treatment will not be compromised in any way if they do not agree to participate or who subsequently withdraw. In line with GDPR guidelines, participant rights to access, change or remove their information will be limited, as we will need to manage information in specific ways in order for the research to be reliable and accurate. If a participant chooses to withdraw from the study, we will keep the information that we have already obtained. To safeguard participant rights, we will use the minimum personally-identifiable information possible at all stages of the study. Study participants will notify the chief investigator and/or lead research team based at King’s College London if they wish to withdraw, using the contact details provided in patient information leaflets for the study and on the website.

2.2.10 Screening for ‘red flags’

Following consent, we will screen patients for 'red flags'. These are potentially serious issues identified by our clinical co-investigators that may be undetected underlying causes of symptoms. After the participant's consent, they will self-complete an online or paper copy checklist (Appendix 4) to check eligibility criteria above. Eligibility will depend on the length of time an issue has been present and whether any action has been taken to report or investigate it:

- If the participant's responses indicate a potentially serious new (within the last 2 weeks) issue that has not been reported/investigated, they will be sent an email or letter explaining that they are currently not eligible, and advising them to seek a consultation with their health care team (Appendix 10a). The email or letter will also provide contact details if they wish to speak to the research team. In addition, if the participant receives their routine IBD care from a hospital that is participating in the IBD-BOOST programme, their IBD team will be notified of the results of their checklist via email or letter (Appendix 10b).

- If a participant's responses indicate a potential medical concern that has been ongoing for 2 weeks to 3 months but has not been reported/investigated, they will still be eligible but will be provided with an email or letter advising them to seek a consultation with their health care team (Appendix 10a). The email or letter will also provide contact details if they wish to speak to the research team. In addition, if the participant receives their routine IBD care from a hospital that is participating in the IBD-BOOST programme, their IBD team will be notified of the results of their checklist via email or letter (Appendix 10b).

- If the potential concern has been present for 3 months or more, this will be considered to be a chronic issue and they will be eligible with no email or letter issued.

- See Appendix 10a for full details of eligibility on "red flags".

Participants and their IBD team are also advised in the email or letter to contact the team if the information provided in the general health assessment has changed. Potential participants who report red flags but are judged to have already had the symptom adequately investigated will be eligible for the RCT. Potential participants who have not had red flags previously investigated will be eligible once they have had these symptoms checked out by their usual care team (IBD team or GP). If the participant's information has changed they can be re-sent the checklist to once again determine their eligibility.

If any clarifications regarding potential 'red flags' are required by the IBD-BOOST team, they will contact the participant using the details provided in the IBD-BOOST Survey.

We will also send consenting participants who have not already taken part in IBD-BOOST Optimise (stage 3 of the programme, who will have already done this) a kit for testing faecal calprotectin level at this point (see below), but non-response to this test will NOT delay randomisation. Participants will be advised of the results of their test via email or letter (Appendix 18a), as will their IBD team for those who receive their routine IBD care from a hospital that is participating in the IBD-BOOST programme (Appendix 18b).

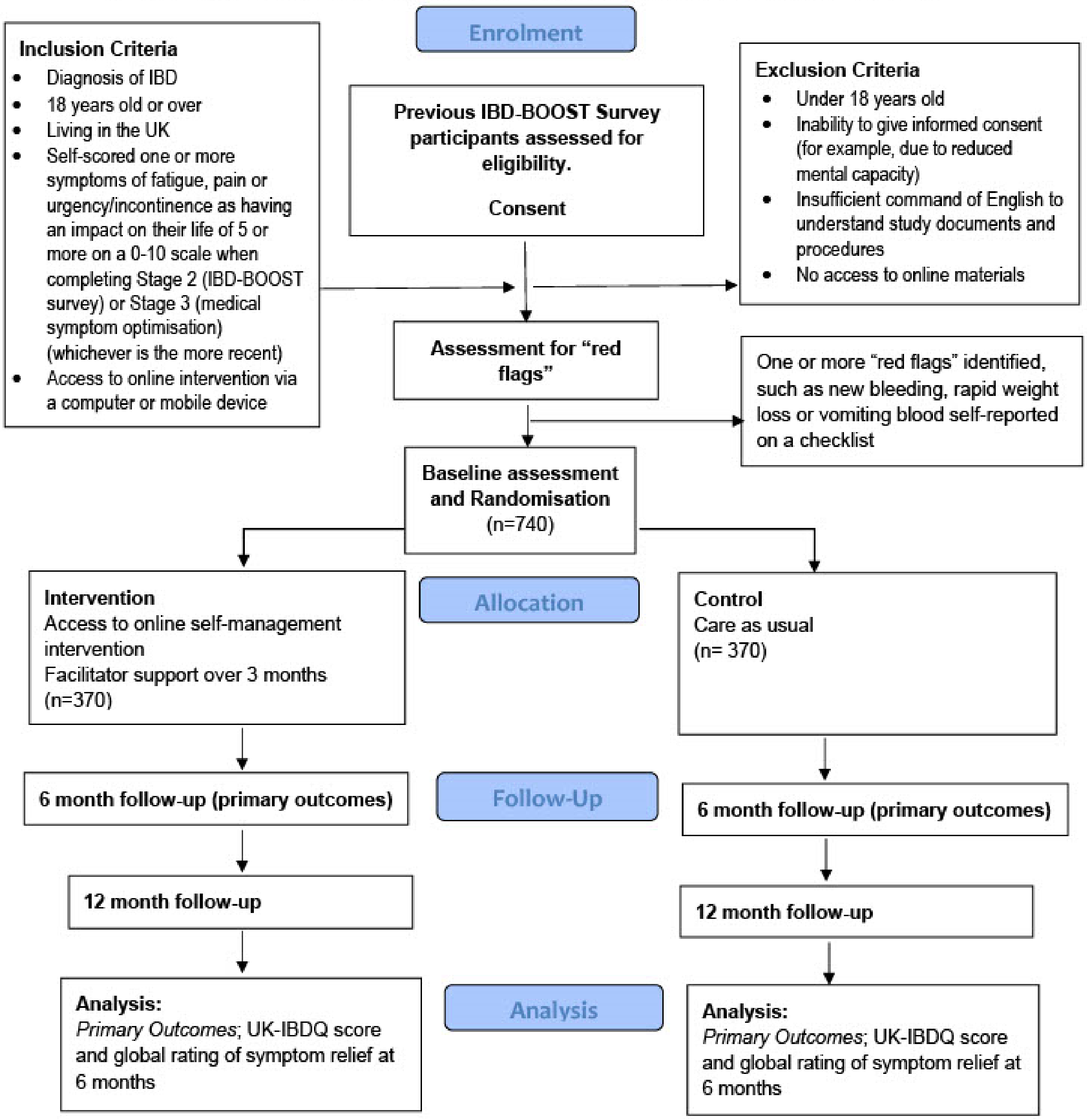
Table 1: IBD-BOOST RCT Screening and data collection schedule

	Completed by	*Database*	*Pre-consent*	*Pre-Baseline*	*baseline*	*6 months*	*12 months*	*Ongoing or during treatment*
Screen for eligibility	TT	KCL	X					
SWAT Randomisation	TT	PCTUD	X					
Invite reply	P	PCTUD	X					
Consent form	P	PCTUD		X				
Red flags assessment	P	PCTUD		X				
Adverse event form	TT	PCTUD						X
Drop-out event form	TT	PCTUD						X

Faecal Calprotectin test	TT	PCTUD		X				
Baseline CRF (Appendix 11)	P	PCTUD			X			
Follow up CRF (Appendix 12)	P	PCTUD				X	X	
Participant Qual Interviews	PET	KCL			X	X		
Intervention Participant PE Questionnaire	PET	KCL				X		
Clinician Qual Interviews	PET	KCL			X	X	X	X
Randomisation	TT	PCTUD			X			
Nurse support	RN	PCTUD						X
Intervention website User Query	TT	PCTUD						X

Key: KCL = King's College London; P = participants; PCTUD = PCTU database; PET = process evaluation team; RN, research nurse; TT, trial team.

Figure 1: Consort diagram for a Randomised Control Trial of supported self-management for symptoms of fatigue, pain and urgency/incontinence in people with inflammatory bowel disease



IBD-BOOST Trial Consort Diagram 5.0 27.05.2022
IRAS ID: 250161

2.2.11 The Intervention

BOOST is a web-based programme for self-management of pain, fatigue and urgency/incontinence symptoms in IBD based on the principles of CBT 'BOOST' was initially mapped out based on the theoretical models described above (see Appendix 13 for logic model) and then developed with extensive patient and clinician engagement. BOOST is currently undergoing extensive testing and was deployed in February 2019 for user feasibility testing prior to the pilot study.

Rationale for web-based delivery: Our previous experience of recruiting people with IBD for interventions (40) and extensive patient engagement and literature on the popularity of electronic self-management in IBD determine our choice of online self-management. Although people want help with symptoms, they do not want repeated hospital visits and may decline help offered on that basis.

The content has been developed to be interactive and tailored to patients' needs. BOOST includes 11 online sessions which can be viewed on computer, smart phone or tablet (however participants are advised that computer or tablet is likely to give a better view of the programme). Sessions 1-7, are core transdiagnostic sessions to be completed by participants experiencing fatigue, pain and faecal urgency/incontinence (See table 2). Based on a Cognitive Behavioural model of IBD symptoms (see Appendix 13), the core sessions cover topics around: understanding IBD symptoms, balancing activity and exercise, sleep hygiene, changing negative thoughts, coping with stress and emotions, and making the most of social support. Sessions 8-11 are symptom-specific sessions to be completed by participants experiencing or with a specific interest in fatigue, pain and faecal urgency/incontinence, respectively (See table 3). The symptom specific sessions provide participants with more in-depth psychoeducation on the interaction between medical and psychosocial factors contributing to the severity and impact of IBD symptoms, together with practical tips and exercises on how to better manage them. See Appendix 14 for a sample text for online session 1: full text for all sessions is available on request.

Participants will be able to log into BOOST via a computer, tablet or smart phone, according to their preference, and complete approximately one session per week. Each session takes 30-60 minutes to complete. Participants will be able to complete the sessions at a time and place that is convenient to them and to pace themselves in a way that suits them and their lifestyle. In between sessions, participants will be asked to complete tasks which will help them to practice the skills acquired during the intervention sessions. Throughout the intervention, the participants will be supported via online messages by an assigned healthcare professional, who will act as their facilitator. The facilitator will have access to information on how participants are using the intervention (sessions and task completed) and will monitor and help to promote participants' engagement with the intervention, together with supporting participants work towards achievement of their intervention goals.

Group 1: Access to care as usual plus the online tailored self-management programme for 6 months, plus one individual telephone or Skype support session (for up to 30 minutes, training and a paper copy of the content will be provided for the facilitator), plus access to online messaging with their facilitator via the BOOST platform for the first 3 months after recruitment.

Group 2: Care as usual (CAU).

All participants will have access to all usual care, including monitoring with clinic visits and/or via the local IBD helpline (this will constitute care as usual for group 2).

Participants in Group 1 with an active IBD flare during intervention will not be expected to discontinue the intervention but will have continued access to the online programme. All randomised participants will be sent the questionnaires to capture outcome measures (Appendix 12) at 6 and 12 months after randomisation, regardless of whether they have incidentally reported a disease flare or not.

Facilitators will be recruited (and trained) from our 15-20 sites, aiming for at least 2 per site, enhancing generalisability. This will mean each facilitator will be offering telephone support to ~11 participants in Group 1 over the duration of the RCT. The programme suggests one 30-minute phone call after participants have completed session 1, and then online messaging from the participant to the facilitator as the participant wishes thereafter. If the participant does not respond after 2 messages sent by the facilitator to schedule the initial call, the facilitator can advise the central study team of this, and they will try to contact the participant outside of the intervention to ensure that they are receiving the BOOST messages. The participant is prompted to consider if they want to send a message at the end of each online session. The facilitator can reply to online messaging within the intervention platform. This is not seen as an unrealistic extra workload.

Participants will log in to the online programme (unique login), undertake symptom self-assessment, prioritise what they want to work on, with signposting to relevant sections depending on symptoms and response to online questions. (Example: link to a section on improving sleep patterns for those with fatigue who nap a lot in the day or with a problem getting to sleep at night).

Table 2: Intervention Core Sessions

Session 1. Understanding your IBD symptoms	• *Welcome to the programme* • *Factors that can contribute to fatigue, pain and urgency in IBD* • *Identifying factors that relate to you* • *Use of self-monitoring not symptom focusing* • *Setting your aims for the programme*
Session 2. Balancing your activity, eating and exercise	• *Why are activity and exercise important?* • *Activity* • *Exercise* • *How fear leads to avoidance* • *Eating patterns* • *Setting your goals for activity and exercise*
Session 3. Improving your sleep	• *Why is sleep important?* • *Sleep habits* • *Sleep patterns* • *Improving your sleep* • *Setting your goals for sleep*
Session 4a. Changing your thoughts: Part 1	• *Why are thoughts important?* • *Identifying unhelpful thinking*
Session 4b. Changing your thoughts: Part 2	• *Developing alternative thoughts* • *Why is managing stress and coping with emotions important?* • *The effects of stress and finding ways to manage it* • *The role of emotions and determining how best to take care of yourself*
Session 5. Managing stress and coping with emotions	• *Why is managing stress and coping with emotions important?* • *The effects of stress and finding ways to manage it* • *The role of emotions and determining how best to take care of yourself* • *Setting your goals for managing stress and emotions*
Session 6. Making the most of your social support and communication	• *Types of social support* • *Communication and disclosure* • *Setting your goals for social support*
Session 7. Summary and maintaining improvement	• *Revisiting your programme aims* • *Preparing for the future* • *Sustaining and building upon improvements* • *Continuing onto the BOOST symptom specific sessions*

Table 3: Intervention Symptom-Specific Sessions

Session 8. Managing and understanding fatigue in IBD	• *Types of fatigue* • *Factors related to IBD fatigue* • *Your vicious cycle of IBD fatigue* • *Practical strategies to manage fatigue*
Session 9. Managing and understanding pain in IBD	• *What is IBD-pain?* • *Acute and chronic pain in IBD: what's the difference?* • *Causes of pain in IBD* • *A model of IBD pain* • *How can I best manage my pain?* • *Common questions around pain in IBD*

Session 10. Managing urgency and leakage	• *Bowel functioning and bowel control difficulties* • *Stress and anxiety in urgency* • *Exercises to help reduce accidents* • *Practical bowel management tips* • *Using social networks to help manage urgency*
Session 11. The role of acceptance and self-compassion in pain	• *What is acceptance?* • *How can acceptance of my pain help me?* • *Role of resilience* • *Practical exercises*

At 12 months, participants allocated to CAU will be offered access to the online intervention (but no facilitator support). Uptake will be monitored. This is clearly stated in the Participant Information Leaflet.

Use of secondary and primary care services and other resources will be collected over the 12-month follow-up period (see Economic Evaluation section below).

We have worked with our patient panel to devise strategies to optimise engagement with the online intervention and retention of our sample, such as personalised text or email reminders.

Facilitator Training and Intervention Fidelity

There will be specific training for facilitators on delivering the intervention and the principles of CBT via a manual (which contains the full text of the online intervention for reference) and a one-day intensive course or online remote sessions delivered by expert psychologists in the intervention development team led by Professor of Health Psychology and lead for intervention development Rona Moss Morris. The training will introduce facilitators to the principles of a cognitive-behavioural approach, practice scenarios to the initial telephone conversation and reviewing the individual participant's model of a vicious circle of symptoms which they will have constructed during session 1 online. The training will also cover responding to online messaging, and what to do in certain situations (e.g. the participant not engaging with the programme, getting stuck, or frequent or persistent use of the messaging service). Adverse events will be discussed and the manual contains standard operating procedures (SOPs) for actions to be taken in certain scenarios, such as the participant appearing very depressed or suicidal). Online remote training sessions will also be carried out to train any new intervention facilitators due to staff turnover.

Facilitators will receive further training and supervision during the site initiation visit from the intervention development/trial team. Prior to starting the trial, facilitators will also receive one-to-one supervision on a 'practice patient' (a person with IBD from our Patient and Public Involvement panel) over a two-week period. A practice telephone session between facilitator and patient will be audio-recorded for training purposes, where facilitators will use the telephone checklist prompt sheet to guide the call. Facilitators will also receive feedback from a member of the intervention development team on the online messaging responses with the practice patient. All facilitator training will be logged via a training log held in the Investigator site file at the respective participating site.

Facilitators will complete the telephone checklist prompt sheet for each trial participant and provide these to the research team, and receive supervision on a trial patient telephone call, upon the patient's permission for a psychologist to listen (but not audio record). Online messaging contacts will be recorded and analysed in the process evaluation (below). Facilitators will receive approximately monthly supervision (or on request of the facilitator) with a trained member of the research team to ensure fidelity and provide support for any difficult issues that may arise. Supervision will include reviewing online messaging and tasks. Participants are told when they commence the online programme that their facilitator and the supervisor will be able to see their messages and tasks.

2.2.12 **Outcome measures**

Trial outcome measures have been confirmed following focus group opinions and refined with our patient panel.

Data collection

Outcome measures will be collected by either direct participant input into a secure online study-developed database managed by the QMUL Pragmatic Clinical Trials Unit (PCTU) or via paper copies sent by central research team at King's College London. Paper copies received in the post will be inputted manually into the database by the research team at King's College London. The questionnaires will not be available in any languages other than English. Each clinic or site will complete a study log for all their participants who have been contacted for the study. Two email or text reminders will be sent to non-responders. If there is still no response an attempt will be made to collect the primary outcome measure by telephone.

Window for return of outcome measures: up to 8 weeks (with reminders) at the 6 month follow-up and up to 4 weeks (with reminders) at the 12 month follow up (with up to 2 text or email reminders: see Appendix 15 for content of reminders). If primary outcome measure is not returned by 8 weeks at 6 months an attempt will be made to have this (the IBDQ only) completed over the phone by a member of the central trial team reading out the questions and recording the response.

Baseline demographic details (age, gender, type of IBD) and ROME IV criteria for irritable bowel syndrome (54) will be collected.

Faecal calprotectin tests and lab processing
We will collect faecal calprotectin results for all participants who have not previously participated in the IBD-BOOST Optimise study as a measure of inflammation (55). Calprotectin results will be obtained by posting sample kits to participants after the baseline questionnaire has been completed. There will be one reminder by email or text for non-responders (Appendix 16). The kits will include a standard and widely used faecal sample pot; packaging and postage will be compliant with 2004 Human Tissue Act. The envelope will have pre-paid postage. The pot sample will be labelled with the participants study ID only. All samples will be processed by King's College Hospital laboratory. No identifying information will be sent to the laboratory at King's College Hospital. All samples will be disposed of following analysis. Results for calprotectin tests on stool samples provided by the participant will be accessed by the central research team on King's College Hospital laboratory's secure results portal and uploaded on to the study database on the Calprotectin CRF (Appendix 17). A copy of the result will be emailed to the participant along with a brief explanation using a template (Appendix 18), advice on further actions that should be taken and contact details for more advice.

Note: return of this sample is NOT required to progress to randomisation.

<u>Data collection tools</u>

Primary outcome measures:

UK Inflammatory Bowel Disease Questionnaire (UK-IBDQ) (56) and global rating of symptom relief (incorporated in Appendix 12) at 6 months after randomisation.

Secondary outcome measures (all measured at baseline, 6 and 12 months after randomisation and considered as outcome measures at 6 and 12 months unless otherwise stated):

- UK-IBDQ (56) at 12 months
- Rating of satisfaction with results of IBD BOOST programme (simple 0-100 visual analogue scale) at 6 and 12 months only
- Global rating of symptom relief at 12 months
- Numerical (0-10) pain rating scale
- Vaizey (faecal) incontinence score (57), reflecting patients' perceptions of severity (58)
- IBD-Fatigue score (59, 60)
- IBD-Control score; 8-item self-reported score to measure disease control from the patient's perspective (61)
- EQ-5D-5L (62) general health-related quality of life

Health economic measure (at baseline and 6 and 12 months after randomisation):

- IBD Resource Use Questionnaire to determine health care and other resource use and costs (see Health Economics section below)

Process evaluation measures: these include measures which may either moderate or mediate the treatment response. These data are important in terms of variables which may help to explain response, to be used as part of process evaluation (see below): (at baseline, 6 and 12 months after randomisation):

- To measure change in general mood: depression as measured by the Patient Health Questionnaire (PHQ-9) (63). This measure includes a question (item-9) that may indicate suicidal thoughts. The action to be taken if suicidal thoughts are indicated is described in Section 5.1 Assessment and Management of Risk below.
- To measure self-efficacy: Self-efficacy for managing chronic disease 6-item scale (64)
- To measure IBD/illness specific cognitions: Brief Illness Perceptions Questionnaire (BIPQ) (65) (minus the open ended causal items). Items are prefaced in relation to beliefs about IBD symptoms rather than IBD generally.

- To detect symptoms which determine a co-diagnosis of Irritable Bowel Syndrome (IBS) and IBD: Rome IV criteria for IBS
- Medications for depression, pain, diarrhoea
- To measure IBD/illness specific affect: Visceral Sensitivity Index (VSI): gastrointestinal symptom-specific anxiety (66)
- To measure behaviour: All-or-nothing and avoidance/resting subscales from the Cognitive and Behavioural Responses to Symptoms Questionnaire (CBRQ)

Note: this outcome set are questionnaires validated for self-completion and will be compiled into a single online questionnaire. We recognise the potential burden on participants of these multiple outcome measures (just under 150 questions). However, as we are assessing multiple symptoms and are keen to include potentially explanatory variables, we feel that these are necessary. We have worked with our PPI representatives on optimising format for outcome measures in a user-friendly online interface. After the pilot study we will review response rates and modify our approach (with an Ethics amendment) if it seems necessary.

The central research team will also send an unconditional £5 inconvenience voucher to participants (Appendix 19 for accompanying letter or email) for both the 6 and 12 months follow-up (see follow-up section 2.2. 16 below).

2.2.13 **Sample size**

The primary outcome UK-IBDQ, ranges from 0 to 120 where low values indicate poor quality of life. Using several published studies (56, 67, 68), we estimate the standard deviation of the change in score to be between 20 and 30. This would mean a standardised effect size of 0.3 would equate to a difference of between 6 and 9 points on the scale. In the validation study (56) the difference in score between those with mild disease and disease in remission was 12 points and the effect of relapse was 10 points.

An effect size of 0.3 was observed in a small study of dietary advice in patients with ulcerative colitis (67) and deemed to be the minimal clinically important difference (MCID). Based on the above considerations, an original sample size calculation estimated that 680 participants would need to be recruited to the trial and the original funding and ethics approval was on this basis. However, during the study the covid-19 pandemic made our anticipated recruitment of 30 facilitators from 20 NHS sites impossible. During team discussions it also became evident that we had not made appropriate statistical adjustments for having two primary end points. The sample size calculation was therefore adjusted before the end of recruitment as follows:

A minimum of 740 participants are going to be randomised, approximately 370 to each group. This allows the MCID difference to be detected with 86.4% power at a 2.5% significance level. It is anticipated that 16 facilitators are participating in the trial. Taking account of a facilitator effect (assuming a facilitator intraclass correlation of 0.04) in the intervention arm, 352 participants are required in each study arm to achieve 86.4%

power (21 participants per facilitator). The sample size is decreased by a deflation factor of 0.84 assuming that baseline values of the outcome measure are predictive of post-treatment values (correlation 0.4) and inflated to account for 20% loss to follow-up resulting in the final recruitment target of 740.

The 20% drop-out assumption is based on drop-out rates from previous studies of self-management: 19.2% of 682 participants in IBD disease self-management (not online) (69) ; 20.4% of 333 participants for online self-management of Ulcerative Colitis disease flares (43); 18% control and 16% intervention of 1140 participants randomised for chronic disease self-management in other diseases (51). Adjustment for correlation between baseline and follow-up values of the primary outcomes is based on Walters et al (70) who suggest a median correlation of a QoL measure with 6 months post randomisation outcome of 0.5 with a lower IQR bound of 0.41. Being conservative a correlation of 0.40 is assumed.

2.2.14 **Randomisation**

Participants who consent, are eligible and return the baseline questionnaire will be randomised by the central research team using an online randomisation system developed for the study by the PCTU. The central team will then inform the participant which group they are in (see Appendix 20 for text and email notification to be sent to participants from the central team informing them of group allocation) and inform the clinical sites of participants in Group 1 who will receive local facilitator support (phone call and online messaging). The local facilitator will be given the participant's details and access to their online tasks.

Stratified central web-based randomisation to 2 groups. Stratified by:

- diagnosis (Crohn's disease vs. other IBD)
- whether or not participated in Stage 3 study (medical symptom optimisation)

2.2.15 **Blinding**

Faecal calprotectin level will be entered into the database by a person blinded to group allocation when the result has been returned from the laboratory and the participant informed of the result. Blinding of participants or facilitators is impossible. The trial steering committee, CI, health economics and statistics teams are blinded, that is, will not see results broken down by treatment arm during the trial or any time prior to analysis plans being signed off. Final analysis will occur once all follow up data is collected, the final statistical analysis plan has been signed off and data cleaning has occurred.

2.2.16 Follow-up

We will adopt evidence-based methods to minimise loss to follow up which have been identified in a systematic review (70). These include providing incentives to participants, contacting respondents prior to sending the follow-up questionnaires and using phone, text and email to contact participants. To optimise our response rates, we will provide an unconditional inconvenience voucher of £5 at 6 months and 12 months by post with a letter or by email (Appendix 19) at the same time as we send out the follow-up questionnaire electronically or by post. Information on incentives will be clearly stated within the participant information leaflet.

2.2.17 Statistical Analysis

Analyses will be performed according to intention-to-treat; all patients with a recorded outcome will be included in the analysis, and analysed according to the treatment to which they were randomised. Summary statistics by group, treatment effects, 95% confidence intervals, and p-values will be presented for primary and secondary outcomes, and process measures. Baseline demographics and outcomes, type of IBD classification, Rome IV IBS and all other follow up data for the two groups will be summarised by treatment group using descriptive statistics, but not subjected to statistical testing.

The two primary outcomes UK-IBDQ and global rating of symptom relief will be compared between the two arms at 6 months after randomisation to assess treatment effectiveness. The significance level for hypothesis testing of the primary outcomes will be Bonferroni adjusted to alpha=0.025 to allow for multiple comparisons.

The primary outcomes will be analysed using mixed model repeated measures analysis with restricted maximum likelihood estimation and an unstructured covariance matrix for residuals. The following covariates will be included in the model as fixed effects: baseline value of outcome measure, stratification factors, fatigue, pain and incontinence at baseline, age and gender. Facilitators will be added as a random effect in the intervention arm only.

Secondary outcomes will be analysed in the same way with inclusion of respective baseline value of respective outcome. Outcomes and covariates may change in the light of new information but will be agreed prior to unblinding of the data.

Pre-specified subgroup analyses to investigate subgroups who might respond better to treatment are described in section 2.2.18.

A sensitivity analysis using imputation methods to allow for missing data under varying assumptions for those lost to follow-up will be carried out.

A detailed statistical analysis plan will be completed and signed off by the Programme Steering Group prior to final database lock.

2.2.18 Planned quantitative process evaluation secondary analysis

The purpose of the quantitative process analysis is to two-fold:

1. To explore moderators of effect of treatment.

This will allow us to assess if there are any key subgroup effects and if the intervention should in future target specific groups of patients. For instance, if depression and disease activity moderate treatment effects, it may be best to focus the intervention on patients in remission and provide alternate treatment to people with depression. If medical treatment optimisation moderates outcome, then again this confirms best practice would be to provide optimised medical treatment before the intervention. Alternately, if there are no significant moderators of effects, the intervention may be generalisable to the broader IBD population.

Subgroup analyses to be investigated are those who underwent Optimise (Stage 3); IBD in remission (faecal calprotectin less than 200 + IBD control <13: need to satisfy both to be defined as in remission); baseline measure of anxiety and depression (PHQ-9 and VSI); ROME IV criteria for IBD met at baseline or not. These will be investigated by adding interactions terms between treatment group indicator subgroup indicator variable to the analysis models used for the primary outcomes (UK- IBDQ and global symptom relief).

2. To explore mediators of change.

Mediation analysis using structural equation models allows us to explore if a treatment effect occurs through hypothesised treatment mechanisms. We have based our choice of mediator measures on the logic model in the Appendix. Putative mediators have been mapped onto the key intervention components i.e. the factors the intervention attempts to target to bring about improvements in symptom and quality of life. These include: the Visceral Sensitivity Index (VSI (66) and PHQ-9 (63) for symptoms of anxiety and depression; Self-efficacy for managing chronic disease 6-item scale (64); Brief Illness Perception questionnaire (65); All-or-nothing and avoidance/resting subscales from the Cognitive and Behavioural Responses to Symptoms questionnaire (CBRQ); IBD-Control (61); faecal calprotectin (55); satisfaction with outcome.

Following best practice, we are looking at whether proximal change in the mediators predicts improvements in the primary outcomes at 12 months.

<u>2.3: Study within a Trial (see protocol in Appendix 3)</u>

Research question:

What is the effectiveness of a brief participant information leaflet (PIL) versus standard length PIL on participant recruitment rates into the IBD BOOST RCT?

Aims & Objectives:

This SWAT aims to evaluate the effectiveness a brief PIL compared with a standard length PIL on recruitment and retention rates in the IBD BOOST RCT.

Methods:

A randomised study within a trial (SWAT) embedded in IBD BOOST. Patients identified as potentially eligible and invited by the central research team to participate in IBD BOOST will be randomised in a 1:1 ratio to be sent the standard length PIL or a brief PIL.

The primary outcome is the proportion of invited participants that are randomised into the IBD-BOOST RCT. Secondary outcome will be retention rate at 6 and 12 months. Outcomes are compared between the brief and standard PIL groups.

Analysis:

Analyses will be on an intention to treat basis, using 2-sided significance tests at the 5% significance level. For the primary outcome, logistic regression will be used to produce odds ratios and their associated 95% confidence intervals and p-values.

2.4: Economic Evaluation

An economic evaluation using recommended methods (71) will be undertaken from an NHS/Personal Social Services and patients' perspectives to examine the cost-effectiveness of the intervention at 12 months. We will estimate the cost of the intervention, including facilitator's contacts with each patient, website hosting and maintenance and facilitators' supervision. The cost of the intervention will also include costs associated with training facilitators (venue, trainer's and facilitators' time, travel and subsistence, administrative support and materials).

Data on the use of health and social care services will be collected using a study-specific IBD Resource Use questionnaire (included into the participant questionnaire: Appendix 11 and 12), developed within this Programme Grant, asking about contacts with primary and secondary care, investigations, medications, hospitalisations, employment and out-of-pocket expenses. Primary care services will include consultations with general practitioners and practice nurses. Medications will include medications for incontinence, pain, fatigue and depression, ostomy supplies and vitamin supplements. Secondary care services will include A&E and specialist visits (including dietician, psychologist and IBD specialist nurses), and IBD-related hospital admissions. Personal out-of-pocket expenses will include over-the-counter medications, cleansing products, meal replacements and alternative therapies. The questionnaire's reliability will have been assessed in a separate study prior to the start of the trial using the test-retest reliability method (separate KCL ethics approval number MRA 18/19-8956) and its online functionality will be tested during the pilot study above.

Unit costs will be sourced from national sources (Personal Social Services Research Unit: Unit Costs of Health and Social Care 2014; PSSRU, 2017; Department of Health: NHS reference costs 2017/18; BNF: British National Formulary 2017); NHS Improvement: National tariff payment system 2017/18) and will be applied to categories of resources used to estimate individual-level total costs. Health-related quality of life data will be collected using the EuroQoL EQ-5D-5L questionnaire administered at baseline, 6 and 12 months.

We will follow the intention-to-treat principle and missing data will be handled using multiple imputation. We will calculate quality-adjusted life years (QALYs) and total costs for each participant in the trial during the 12-month follow-up. We will evaluate the additional QALYs and costs with allocation to the BOOST intervention and the intervention's cost-effectiveness.

An incremental cost-effectiveness ratio (ICER) will be estimated as the additional cost per additional QALY gained. Uncertainty around the ICER point estimate will be assessed (72). The probability of the online IBD self-management intervention being cost-effective compared to usual care will be estimated at the NICE threshold of £20,000-30,000 per QALY gained. One-way sensitivity analyses will be conducted to address the uncertainty associated with the cost of the intervention and healthcare services use.

For generic quality of life and resource use outcomes we will consider adjusting in the analysis for baseline values of respective outcomes: e.g. EQ-5D utility at baseline and level of costs in the previous 3 months.

In a sub-set of up to 4 intervention sites research nurses will also complete the IBD Resource Use Questionnaire using hospital records (NHS costs sections, excluding the personal costs: Appendix 11) to assess the accuracy of patient completed measures (consent for access to medical records is included in the Trial Consent Form). Paired comparisons will be made between patient completed and research nurse completed data and correlation reported.

2.5. Process evaluation

Research question

What factors (promoters and blockers, individual and organisational) impact on the completion and effect of the intervention?

A mixed methods process evaluation of the RCT will be conducted based on the MRC guidance (73). The design of the process evaluation is based upon the logic model (Appendix 13) and consequent discussion within the study team, so that the process evaluation can explore the areas of greatest uncertainty.

Aims

To investigate the processes through which the intervention is delivered, and what is actually delivered in practice, to aid the interpretation of the results of the main trial and to inform future rollout and implementation of the intervention, if successful.

Objectives are to:

- Determine expectations at baseline through interviews
- Determine if the intervention was delivered as planned (intervention fidelity).
- Explore participant responses to the intervention both quantitatively (through routinely collected study process data around online log-ins and programme usage and the COVID-19 survey) and qualitatively (through interviews and the COVID-19 survey)
- Explore participants' responses to being in the control arm of the study through interviews
- Understand IBD nurses' views of the intervention and its integration and usability in everyday NHS care through interviews
- Consider any potential contextual influences on the intervention implementation and outcomes, including a survey at 6 months to look at the impact of COVID-19 upon engagement with the intervention.

Based on the logic model developed for the intervention (Appendix 13), which provides a detailed description of the intervention and its causal assumptions, we will seek to monitor intervention fidelity and provide insights into how the intervention did or did not work in practice, any unintended consequences, as well as providing information to aid future implementation and dissemination. A list of key assumptions and uncertainties has been developed by the intervention development team, following completion of intervention development. These will be explored and tested in this process evaluation.

The process evaluation will run concurrently with the RCT in a largely “passive” model (73). However, this passive model will not be operational during our internal pilot study and the teams will work closely together during the pilot to address any issues raised.

Adherence:

- Fidelity to the protocol: proportion of people randomised to the intervention who clicked on the link to the online intervention and commenced the intervention and completed a minimum of four online sessions will be considered to have adhered to the intervention. We will also assess the date the baseline CRF is completed, date “red flags” were completed, and the date randomised to check study processes worked as intended.

Quantitative:

- Monitoring on-line log-ins (number and spacing/timing of log-ins and time spent on each section and in total): automatically collected by the programme software, enabling us to see how participants interact with the intervention.
- A record of number and time of facilitator telephone support sessions or emails (Appendix 21 & Appendix 22).
- Section 2.2.18 gives details of quantitative analysis of possible moderators and mediators of response to intervention.
- All participants allocated the intervention will also receive an online survey at 6 months to look at the impact of COVID-19 on their engagement with the programme.

Qualitative:

- Fidelity to the intended facilitator support: a sub-set of facilitators’ messages will be analysed and compared with instructions in the training manual for fidelity to the intended facilitator support. All facilitator interactions online with participants will be captured and randomly selected interactions (messaging) will be stripped of all personal identifying data and subjected to content analysis (74), to enable assessment of intervention fidelity (as defined by the training day and intervention manual).
- Face-to-face, telephone or Skype interviews with a purposive sample of up to 30 recruits (or until apparent data saturation) before and after the intervention (attempting to recruit the same people before they know their group allocation and then the same people 6 months later; if it is not possible to recruit people for a second interview at 6 months (or when they drop out if this is sooner), these participants will be replaced by another participant who was allocated to the same group with broadly the same characteristics). Interviews will be completed to understand expectations and experiences of the intervention, its acceptability and which aspects they felt where most or least helpful (informing adjustment to the intervention before future roll-out), and for their opinions on changes in their cognitions and behaviours. All participants allocated the intervention will also receive an online survey at 6 months to look at the impact of COVID-19 on their engagement with the programme. We will also interview at least two thirds of the facilitators who are IBD nurse specialists (approx. 20: or until apparent data saturation) to understand their views on supporting the online intervention and fit with their existing workload, both prior to starting patient facilitation and before they finish facilitating the intervention.

  Participants who have indicated on the trial consent form that they are happy to be approached about interviews will be sent a participant information leaflet for the interviews (Appendix 23). Those indicating that they are willing will be purposively selected according to a sampling frame. Facilitators will be contacted by the central trial team using the staff information leaflet (Appendix 24); those indicating willingness will be interviewed. The topic guides for interviews (Appendix 25) have been developed with our PPI panel, some of whom will also be trained to assist with analysis and interpretation of the data. The data will be analysed iteratively and as the interviews progress the topic

guide will be adapted, based on themes which emerge from earlier interviews, to enable exploration of issues which appear relevant in later interviews.
The interview sample will be purposively selected to include both sexes and allocation groups, a range of ages, both IBD diagnoses, and differing numbers of symptoms to better understand patient perspectives on the intervention, experiences of being in a wait list control and whether this altered behaviours and cognitions around symptoms, whether non-completion was due to the design and demands of the intervention, or due to other factors.as well as any unanticipated pathways and consequences. All interviewees will be asked to sign a consent form (face to face) or the questions will be read out by the interviewer (telephone or Skype interviews) and agreement to each statement will be audio recorded. Appendix 26 and 27 gives the consent forms for interviews for participants.

- Analysis: Interpretive data analysis will be informed by the Analytical Hierarchy Framework (AHF) (74), guiding methods for handling, analysing and generating findings from qualitative data. This generic framework is appropriate for exploratory qualitative work as it guides the process of analysis through basic organisational data management stages, to descriptive and finally interpretive levels. The AHF acts as a set of instructions for progression through analysis; the specific detailed method of analysis is achieved through simultaneous use of a specific coding frame as part of that process.
- We will train patient volunteers to co-analyse anonymised qualitative data (as we have done successfully in other studies (4, 75-78), and help us interpret and present our findings widely to patient and media audiences. Training days will be delivered by our qualitative researchers. Training involves an overview of the purpose of analysis, a PowerPoint demonstration of moving from raw data to identification of themes, and a practical session working on existing anonymised data. A handout covering the key aspects is provided, giving people reference material after the training. PPI members and the qualitative researchers all carry out an initial analysis of the data using a coding frame based on the topic guide. Early findings will be collated, with new items identified in the first-round analysis being added to the coding frame. The full team (PPI and researchers then conduct a second analysis using the enhanced coding frame. This iterative process can be repeated as required to ensure a thorough and robust analysis of the data.
- NVivo software will be used to manage data and enable sorting, labelling and retrieval of data segments prior to the human endeavour of interpretation and representation of findings.

The process evaluation team will work to integrate these sources of qualitative and quantitative data into a coherent report which seeks to illuminate the results of the RCT. There will also be a mediation analysis. This will then be presented to the Programme Steering Group for detailed discussion to inform dissemination.

RELEVANT FOR ALL STAGES:

3. <u>Data management</u>

Live data will be stored on a secure database held by the Pragmatic Clinical Trials Unit (PCTU), Queen Mary University of London. Participants will complete the outcome measures via an online link and will input their data directly into the database or complete paper copies and send via a provided stamped addressed envelope to King's College London to be held securely and inputted in the online PCTU database.

Paper copy data and qualitative data will be stored securely at KCL for 10 years in accordance with the organisation's long-term Archiving Standard Operating Procedures. All interviews will be digitally recorded, anonymised, professionally transcribed verbatim and analysed using, if appropriate, NVivo8 software for data management. The interview data will be analysed by the qualitative researchers and PPI representatives independently in researcher's offices and at the University sites (King's College London & QMUL PCTU). Anonymised transcripts will be stored on a password protected computer at King's College London Audio files will be sent via the KCL secure file transfer service to a KCL approved transcriber who will delete their copy of the audio file and the electronic transcription once the latter has been returned to the research team. The transcriber will remove all identifiable data and render these anonymous. Participants are advised of this in the Participant Information Leaflet. All original and transcribed interview data files will be password protected on computers (e-copies) and physically filed (hard copies) in a locked cabinet at the University site.

Audio files of interviews will be deleted at the end of the programme (planned for November 2023) and confirmation of destruction will be recorded. All other data arising from this study electronic or paper format will be kept for a period of ten years for data management governance as required by the sponsor.

3.1 Data protection and confidentiality of participants

The online outcome measures will be administered on the REDCAP system. QMUL BCC IT Security is responsible for the security of the QMUL REDCAP service. Any data entered is securely stored in the PCTU safe haven (BCC) in their Enterprise level data centres. Data will be backed up daily. Access to the system for data entry staff requires a user account which will be issued and controlled by the PCTU Data Management Team. All data analyses will be conducted using only the unique study ID. Transfer of data will occur through strong encryption.

Data collected may be used to support other research in the future and may be shared anonymously with other researchers as stated in the PIL and consent form subject to the necessary regulatory approvals being place.

4. Ethics

4.1 Ethics favourable opinion, HRA approval and NHS R&D

The Chief Investigator will obtain approval from a recognised NRES Research Ethics Committee & Health Research Authority (HRA) or the corresponding bodies within the devolved participating nations. The study must be submitted for assessment of capacity and capability at each participating NHS Trust using the HRA/REC approved study document set and Statement of Activities (SoA) & Schedule of Events (SoE). The Chief Investigator will require a copy of each participating site's confirmation of capacity and capability, agreed statement of activities, PI & research teams current CV and GCP certificates, before accepting participants into the study. This information will be stored in the Main Trial file located at the research team's offices at King's College London. The study will be conducted in accordance with the recommendations for physicians involved in research on human subjects adopted by the 18th World Medical Assembly, Helsinki 1964 (incl. later revisions) and any other relevant ethical guidance. For any Non-NHS sites, a Site-Specific Assessment Form (SSI) will need to be submitted for REC review. All participating sites will require confirmation to commence with recruitment from the central research team at King's College London before the trial can begin at a site.

Amendments

On obtaining a favourable ethical opinion and HRA approval any subsequent changes to the study conduct, design or management will be notified to the original approving REC & HRA and any other relevant regulatory authority via the UK Amendment process (http://www.hra.nhs.uk/research-community/during-your-research-project/amendments/) Authorisation will be sought from the study sponsor for any future substantial and non-substantial amendments arising during the course of the study, prior to submission to the relevant Research Ethics Committee & HRA.. Changes to the study will not be implemented until REC/HRA approval has been obtained unless the clinical need warrants this, for example when urgent safety measures are required. The Chief Investigator or designee will work with sites (R&D departments at NHS sites as well as the study delivery team) so they can put the necessary arrangements in place to implement the amendment to confirm their support for the study as amended. NHS R&D Amendment continuation of capacity and capability will be sought (where applicable) from the participating sites before any substantial changes can be implemented at the applicable site. Details of consent procedures are addressed above. An annual progress report will be submitted to the sponsor and the approving REC/HRA by the CI.

5. Study Conduct Responsibilities

5.1 Assessment and management of risk

This is a low risk study (as assessed by PCTU), although there is potential for participants to become distressed when thinking about their symptoms. The intervention site will include a link / website address to CCUK who provide support via their helpline, and contact details are included in the Participant Information Leaflet. The outcome measures include questions on anxiety and depression. At the end of the questionnaire there are helplines listed that can offer support.

The baseline CRF and 6-month and 12-month CRFs include the PHQ-9, a measure of depression that includes a question which may indicate suicidal thoughts (Item 9: 'Thoughts that you would be better off dead or of hurting yourself in some way'). There are four possible responses indicating the frequency of these thoughts in the past two weeks. If a participant checks the boxes of the two most frequent occurrences of these thoughts

(‘More than half the days/nearly every day’) at baseline, or checks these at 6 or 12 month follow up when they were not checked at baseline, the participant will be contacted by the central research team within ten working days and a PHQ-9 risk assessment undertaken (see Appendix 29). If the same boxes that resulted in a risk assessment at baseline are checked again at 6 and 12 months, the risk assessment will not be repeated: the risk assessment will be done at 6 or 12 months only for a new report on this item. We have piloted this process in a preliminary feasibility study of the IBD-BOOST intervention for pain and it has worked well.

The study will be entered for adoption on to the National Institute for Health (NIHR) Portfolio. All participating sites will be required to follow any applicable procedures outlined by the NIHR and PIs are responsible for uploading recruitment figures in the specified recruitment reports to the Chief Investigator’s designated Data Support Coordinator.

5.2 Safety

Note: due to the nature of the trial, with no face to face contact with participants in either group and minimal contact with the facilitator in the intervention group only, it will not be possible to closely monitor adverse events. Our primary way of capturing adverse events is using responses to safety questions in the 6 and 12 month follow up questionnaire, and these will include items around COVID-19. We consider that the trial carries a very low risk of adverse events. Adverse events may also be recorded by facilitators through communication with participants in the intervention arm. These will be reported using the sponsors form. We will also attempt to capture any unforeseen consequences in the qualitative interviews.

Adverse events (AEs): AE are any clinical change, disease or disorder experienced by the participant during their participation in the trial, whether or not considered related to the use of treatments being studied in the trial.

Serious adverse events: An AE is defined as serious (an SAE) if it results in one of the following outcomes (captured in the participant outcome questionnaire Appendix 12, at 6 and 12 months):

- A life-threatening AE
- In-patient hospitalisation or prolonged hospitalisation not related to IBD flares, which are expected events
- Persistent or significant disability/incapacity
- A congenital anomaly/birth defect in the offspring of a subject
- Is otherwise considered medically significant by the investigator
- Other medical events requiring intervention to prevent one of the above outcomes.

Follow-up after SAEs: An SAE occurring to a research participant should be reported to the main REC where in the opinion of the Chief Investigator (CI) the event was:

- Related – that is, it resulted from administration of any research procedures and
- Unexpected – that is the type of event is not listed in the protocol as an expected occurrence

The chief investigator or sponsor will complete and send a SAE report for non-ctimps (clinical trial of investigational medical products) to the REC within 15 days of becoming aware of the event.

After a SAE, a decision will be made by the trial team, after advice from the relevant authorities and the participant’s IBD team, as to whether the participant should be withdrawn from either their randomised treatment or from the trial. However, we do not envisage a situation, except death, in which a participant would need to be withdrawn.

Arrangements will be made by the trial team for further assessment and management as agreed with the relevant authorities, GP and participant.

The investigator will provide the trial team with a 1-month follow-up report on all SAEs. Further monthly reports should be provided in the absence of resolution. These reports will be communicated to the Trial Steering Committee, REC, and to the local R&D office. Blank Adverse Event Forms will be distributed to sites that are recruiting.

SAEs that do not require reporting: Expected AEs include planned/elective hospitalisations, or unplanned but expected hospitalisation due to flare-up of IBD: these are expected during the course of the trial and will not be collected as SAEs.

Stopping the trial

There is no Data Monitoring Committee for this low risk study. The trial may be prematurely discontinued by the Sponsor or Chief Investigator on the basis of new safety information or for other reasons given by the Trial Steering Committee or REC concerned. The trial may also be prematurely discontinued due to lack of recruitment or on advice from the Programme Steering Committee (if applicable), who will advise on whether to continue or discontinue the study and make a recommendation to the sponsor. There will be no formal stopping rules based on the intervention outcomes. In the unlikely event that the study is prematurely discontinued, active participants will be informed and no further participant data will be collected.

5.3 Confidentiality

The Chief Investigator will preserve the confidentiality of participants taking part in the study and will work in accordance with the Caldicott Principles, Data Protection Act 2018, NHS Code of Confidentiality and any relevant NHS Trust organisational policies or other applicable Data Protection legislation. Participating NHS sites will be bound to act in accordance with these applicable regulations.

5.4 Indemnity

The Study is sponsored by the LNWUH NHS Trust; the NHSLA Indemnity scheme will cover the study.

5.5 Sponsor

LNWUH NHS Trust will act as the main sponsor for this study and will ensure that the trial adheres to the UK Policy Framework for Health &Social Care Research 2017), and any amendments or subsequent replacements. Delegated responsibilities will be assigned to the NHS Trusts taking part in this study via study site and service level agreements.

5.6 Funding

The study is funded by the NIHR Programme Grants for Applied Research funding stream.

5.7 Audits and inspections

The PCTU quality assurance manager will conduct a study risk assessment in collaboration with the CI. Based on the risk assessment, an appropriate study monitoring and auditing plan will be produced according to PCTU SOPs. This audit plan will be authorised by the C.I before implementation. Any changes to the audit plan must be agreed by the PCTU QA manager and the C.I. The risk assessment for the trial will be under continual review to assess relevance, applicability and to identify any actions that may be required.

The study may be subject to inspection and audit by LNWUH NHS Trust under their remit as sponsor and other regulatory bodies to ensure protocol compliance and adherence to GCP and the UK Policy Framework for Health & Social Care Research 2017).

Participating NHS sites are required to comply with any requests for audit by the PCTU or the sponsor or applicable site R&D Office and ensure that the study documentation and information is available. Copies of audit / monitoring reports should be sent to the Sponsor R&D office. Protocol deviations, non-compliances, or breaches from the approved protocol must be reported to the Sponsor R&D Office and PCTU within 24 hours of becoming aware of the event.

5.8 Intellectual Property

Any intellectual property arising from the development, conduct and completion of this study will be owned by the study sponsor.

6. Study Management

The day-to-day management of the study will be co-ordinated through the IBD-BOOST programme manager in collaboration with the CI, and through the Project Management Group.

The Programme Management Group deliberates the practical and logistical aspects of the study, for example, agreeing local procedures for recruiting, including who will recruit, who will consent, workspace available for recruiting / consenting and how this will operate alongside standard clinic operations. The programme is overseen by a Programme Steering Committee whose members are independent of the programme and liaise with the study funders, the NIHR. In the case of study deviations or serious breaches of protocol, study deviation forms will be completed and forwarded to the programme steering committee and the study sponsor.

6.1 Study timelines

We plan to start recruitment on 1st December 2019 and to recruit a minimum of 740 patients over a period of 31 months ending on 31st July 2022; this is approx. 23 patients a month. However, we anticipate that recruitment will be slower in the first few months owing to time taken to set up new sites and that some patients will need to go through the Optimisation process before randomisation.

The first 100 patients randomised or the first twelve months (whichever is sooner) will constitute the pilot phase after which we will compile a report for the PSC. We anticipate the PSC will meet (possibly virtually) following this to make a decision on the future of the trial and any changes needed.

See Appendix 28 for study GANTT chart. A more detailed version will be used by the programme management group to regularly monitor milestones and recruitment.

7. Dissemination

The Sponsor owns the data arising from the study and the CI will act as the custodian of this data on behalf of the sponsor. On completion of the study, the data will be analysed and tabulated and a Final Study Report prepared. The full study report can be accessed on request to the CI. The participating investigators will have rights to publish any of the study data by prior agreement with the CI and the Programme Management Group. There will be no time limits or specific review requirements on the publications.

The funding body (NIHR) will be acknowledged within the publications and review draft publications before journal submission. Participants can request copies of a lay summary of the final report of the outcome of the study by indicating that they want this and giving permission to store contact details for this purpose on the consent form. We do not plan for the study protocol, full study report, or anonymised participant level dataset to be made publicly available.

7.1 Publication Policy

To health professionals who could develop services, and the academic community
We will submit results for publication in multidisciplinary academic journals (such as Inflammatory Bowel Diseases and Journal of Crohn's & Colitis) to disseminate to professional audiences. We will submit to key IBD conferences, including the UK British Society of Gastroenterology, the European Crohn's & Colitis Organisation and the USA Digestive Diseases Week.

Anticipated professional publications:

- Main results paper
- SWAT paper
- Health Economics paper
- Process Evaluation paper
- Study Protocol

To patient groups
We will work with our patient and public volunteers, training those who are willing to present results at local and regional Crohn's & Colitis UK meetings. We will alongside patients to construct a user-friendly lay summary for the CCUK newsletter and website. We will prepare a more detailed summary of results in lay

language for participants and people with IBD who request this and adapt the charity information sheet on bowel control, fatigue and pain accordingly. We will discuss dissemination via their newsletter with the European Federation of Crohn's & Colitis (patient) Associations. The study team are members of all these groups.

Authorship eligibility guidelines and any intended use of professional writers

The co-applicants and anyone else who makes a substantive contribution during data analysis or interpretation (including those involved through PPI) will be eligible to be co-authors.

The results from different centres will be analysed together and published as soon as possible. All publications will follow the guidance set out by the NIHR in 'Identity guidelines and research outputs management for principal investigators'. Individual clinicians must not publish data concerning their patients that are directly relevant to questions posed by the study until the Trial Management Group has published its report. The Trial Management Group will form the basis of the Writing Committee and advice on the nature of publications.

All publications shall include a list of participating centres, and if there are named authors, these should include the project's Chief Investigator. The order of the named authors should be agreed in advance and should reflect the applicable level of input by each author.

No verbal or written report may be made without the approval of the Project Management Group. The sponsor should be consulted for review of any final report before it is disseminated for wider circulation or publication.

8. Peer Review

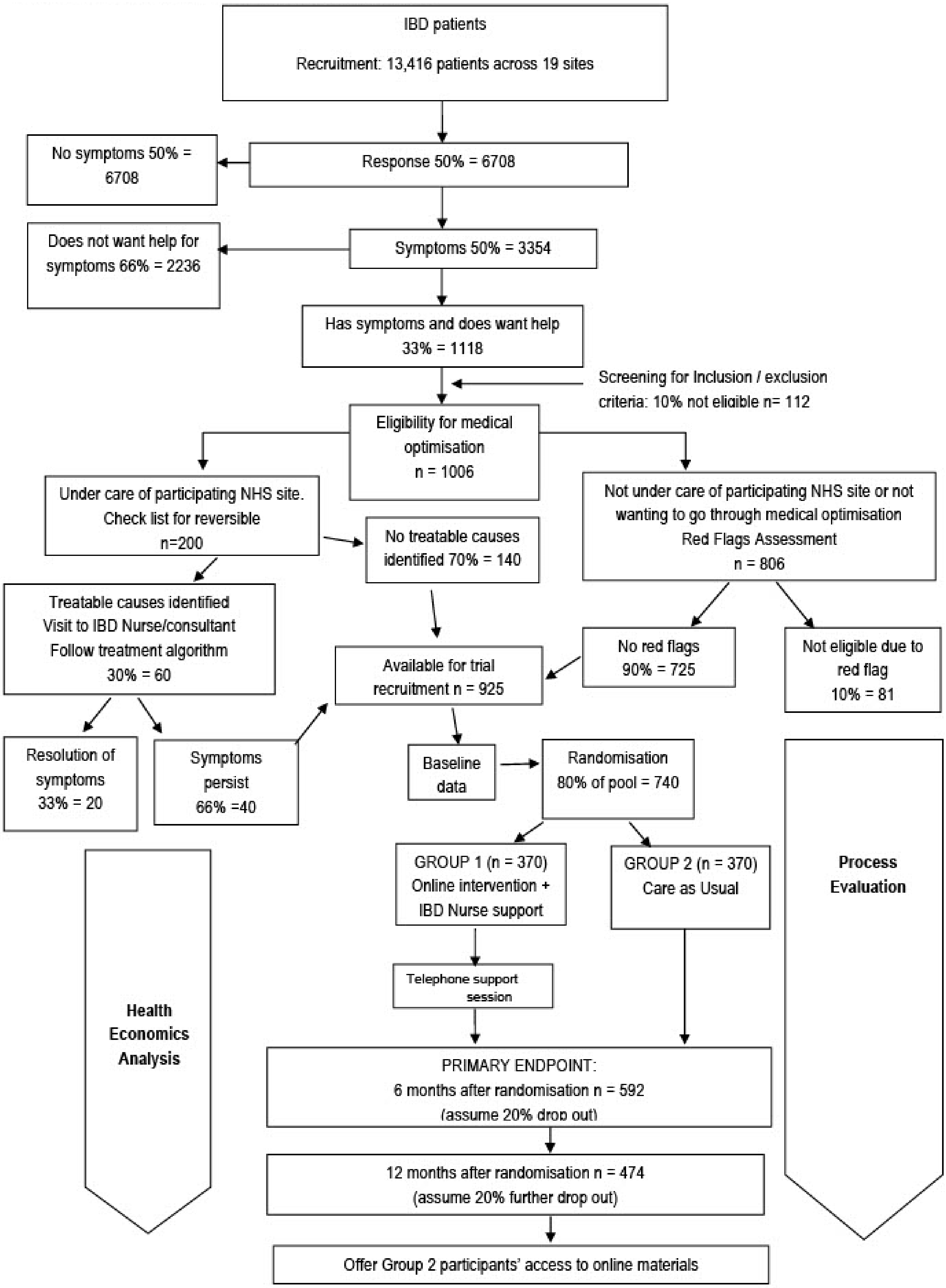
15 peer reviewers commented as part of grant funding process: 2 stage grant application with amendments made as suggested by peer reviewers.

9. Patient & Public Involvement

People with IBD have been extensively involved in developing this research. In particular PPI has informed or will inform:

- Identification of the research questions for the programme
- Development of the intervention
- Design of the research (including development of patient-facing materials)
- Management of the research
- Undertaking the research
- Analysis of results
- Dissemination of findings

Appendix 1: IBD BOOST Programme Participant Flow Chart



Appendix 2: PROTOCOL VERSIONS

Version Stage	Versions No	Version Date	Appendix No. Detail the reason(s) for the protocol update
Previous	1.0	01/04/2019	N/A
Previous	2.0	09/08/2019	• Information on use of paper copies has been added to pages 15, 16, 22, 25 and 31 • We have added 5a,b and c into the appendix for standard PiLS to list paper copies for participants coming through the two different routes (via the IBD-BOOST Optimise study or directly from the IBD-BOOST survey). • We have corrected an error in 2.2.2 Secondary research questions, question (5). We had defined active disease incorrectly and have removed. • We have corrected an error on 2.2.14 Randomisation, The last line 'Allocation will be concealed until consent and baseline measurements completed' was in correct and has been deleted. • We have corrected the study timelines to be in-line with what was originally stated on our IRAS document.
Previous	3.0	16.06.2020	• Added REC reference number • Updated information on reminders • Information on red flags has been updated • Updated data collection schedule • Updated information about scheduling the initial 30 minute phone call • Updated information about facilitator training and intervention fidelity • Updated information about £5 unconditional inconvenience payment, which will now be a £5 voucher • Added information about process evaluation and the process evaluation survey • Updated safety section with note about COVID-19 questions • Updated information about the pilot and PSC meeting after • The list of appendices has been updated
Previous	4.0	08.04.2021	• Covid-related amendment to enhance recruitment: Extension of proposed

			recruitment end date to 30th June 2022, and of total study duration to 42 months. • Change of proposed recruitment end date to 30th June 2022, and of total study duration to 42 months. Change of proposed follow-up end to 30th June 2023, and end of the IBD-BOOST Programme to November 2023. • Update of data management, study timelines and Appendix 28; IBD-BOOST TRIAL GANTT chart to reflect the changes to recruitment, follow-up and programme end dates. • Sponsor representative details and programme co-investigators, research team and programme steering committee members updated.
Previous	5.0	28.05.2021	• Update of IBD-BOOST RCT Screening and data collection schedule to clarify timescale for follow-up facilitator process evaluation interviews. • Update of Facilitator Training and Intervention Fidelity to clarify that training can be via a one-day intensive course or online remote sessions, and that online remote training sessions will also be carried out to train any new intervention facilitators due to staff turnover, rather than a video version of the training. • Update of process evaluation qualitative section to include that facilitators will be interviewed both prior to starting patient facilitation and before they finish facilitating the intervention. Also to clarify that participants will be purposively selected to include both sexes and allocation groups, a range of ages, both IBD diagnoses, and differing numbers of symptoms, but not whether they did or did not complete. • Update of safety to clarify SAEs that do not require reporting rather than AEs.
Current	6.0	21.03.2022	• Update of sponsor representative and trial statistician contact details. • Update of research sites. • Extension of proposed recruitment end date to 31st July 2022, update of number of participants to at least 740, and clarification of methodology and analysis. • Programme co-investigators, research team and programme steering committee members updated. • Clarification of study design and primary outcomes. • Update of Figure 1: Consort diagram for a Randomised Control Trial of supported self-management for symptoms of fatigue, pain and urgency/incontinence in people with inflammatory bowel disease to amend number of participants.

			• Clarification of secondary outcome measures. • Update of sample size and power calculation, and update of corresponding references. • Clarification of randomisation stratification and blinding procedures. • Update and clarification of statistical analysis. • Clarification of Study Within a Trial methods. • Update of study timelines with revised number of participants, recruitment period and recruitment end date. • Update of Appendix 1: IBD BOOST Programme Participant Flow Chart.

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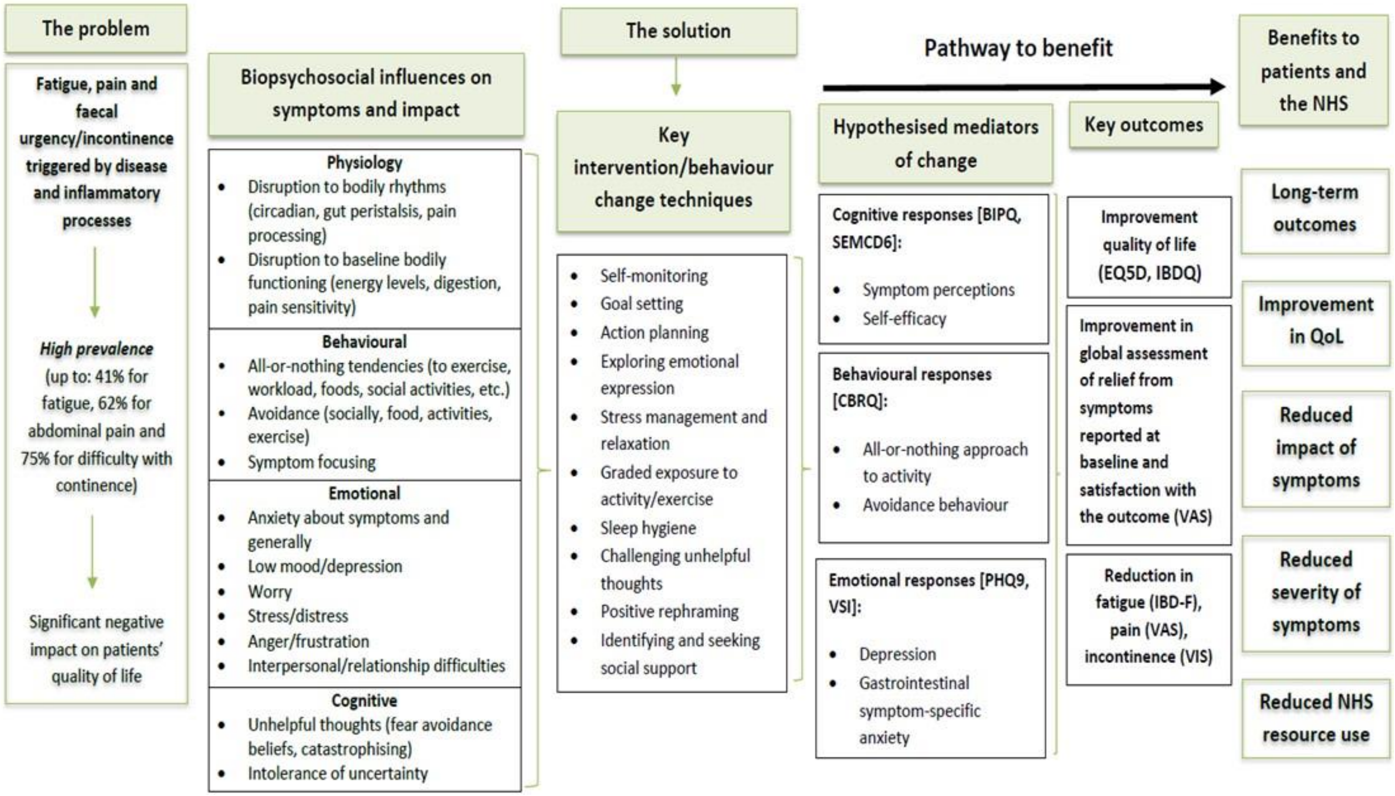
Online supplementary material 2: – Logic model developed for the IBD-BOOST intervention



Online supplementary material 3: Session content for the IBD-BOOST intervention

Core sessions	
1. **Understanding your IBD symptoms** *Tasks: complete individualised vicious cycle + symptom diary*	• Welcome to the programme • Factors that can contribute to fatigue, pain and urgency in IBD • Identifying factors that relate to you • Use of self-monitoring not symptom focusing • Setting your aims for the programme
2. **Balancing your activity, eating and exercise** *Tasks: reviewing goals for activity + sleep diary*	• Why are activity and exercise important? • Activity • Exercise • How fear leads to avoidance • Eating patterns • Setting your goals for activity and exercise
3. **Improving your sleep** *Task: reviewing goals for sleep*	• Why is sleep important? • Sleep habits • Sleep patterns • Improving your sleep • Setting your goals for sleep
4. **Session 4a: Changing your thoughts: Part 1** *Task: thought record*	• Why are thoughts important? • Identifying unhelpful thinking
5. **Session 4b: Changing your thoughts: Part 2** *Task: alternative thought record*	• Developing alternative thoughts
6. **Session 5: Managing stress and coping with emotions** *Tasks: reviewing goals for stress + stress diary*	• Why is managing stress and coping with emotions important? • The effects of stress and finding ways to manage it • The role of emotions and determining how best to take care of yourself • Setting your goals for managing stress and emotions
7. **Session 6: Making the most of your social support and communication** *Task: reviewing goals for social support*	• Types of social support • Communication and disclosure • Setting your goals for social support
Symptom-specific sessions	
8. **Session 7: Understanding fatigue in IBD**	• Types of fatigue • Factors related to IBD fatigue • Your vicious cycle of IBD fatigue • Practical strategies to manage fatigue
9. **Session 8: Understanding pain in IBD**	• What is IBD-pain? • Acute and chronic pain in IBD: what's the difference? • Causes of pain in IBD • A model of IBD pain • How can I best manage my pain? • Common questions around pain in IBD
10. **Session 9: Managing urgency and leakage**	• Bowel functioning and bowel control difficulties • Stress and anxiety in urgency

	• Exercises to help reduce accidents • Practical bowel management tips • Using social networks to help manage urgency
11. **Session 10: The role of acceptance and self-compassion in pain**	• What is acceptance and how can it help me? • Role of resilience • Practical exercises
Summary session	
12. **Session 11: Summary and maintaining improvement**	• Revisiting your programme aims • Preparing for the future • Sustaining and building upon improvements

Note: participants were advised to complete sessions 1-7 (core sessions) and then choose which symptom-specific sessions were relevant to them, finishing with session 12 on maintaining improvement. In user testing each session took approximately 45 minutes to complete.

Online supplementary material 4: Consort checklist for the RCT

CONSORT 2010 checklist of information to include when reporting a randomised trial*

Section/Topic	Item No	Checklist item	Reported on p
Title and abstract			
	1a	Identification as a randomised trial in the title	1
	1b	Structured summary of trial design, methods, results, and conclusions (for specific guidance see CONSORT for abstracts)	Abstract 5-6
Introduction			
Background and objectives	2a	Scientific background and explanation of rationale	Background 7-8
	2b	Specific objectives or hypotheses	8
Methods			
Trial design	3a	Description of trial design (such as parallel, factorial) including allocation ratio	8-9
	3b	Important changes to methods after trial commencement (such as eligibility criteria), with reasons	n/a
Participants	4a	Eligibility criteria for participants	9
	4b	Settings and locations where the data were collected	12
Interventions	5	The interventions for each group with sufficient details to allow replication, including how and when they were actually administered	10-11 + published pa
Outcomes	6a	Completely defined pre-specified primary and secondary outcome measures, including how and when they were assessed	11-12
	6b	Any changes to trial outcomes after the trial commenced, with reasons	n/a
Sample size	7a	How sample size was determined	9
	7b	When applicable, explanation of any interim analyses and stopping guidelines	None: stated
Randomisation:			
Sequence generation	8a	Method used to generate the random allocation sequence	9
	8b	Type of randomisation; details of any restriction (such as blocking and block size)	9
Allocation concealment mechanism	9	Mechanism used to implement the random allocation sequence (such as sequentially numbered containers), describing any steps taken to conceal the sequence until interventions were assigned	9
Implementation	10	Who generated the random allocation sequence, who enrolled participants, and who assigned participants to interventions	10

Blinding	11a	If done, who was blinded after assignment to interventions (for example, participants, care providers, those assessing outcomes) and how	10
	11b	If relevant, description of the similarity of interventions	N/A
Statistical methods	12a	Statistical methods used to compare groups for primary and secondary outcomes	12-15
	12b	Methods for additional analyses, such as subgroup analyses and adjusted analyses	12-15
Results			
Participant flow (a diagram is strongly recommended)	13a	For each group, the numbers of participants who were randomly assigned, received intended treatment, and were analysed for the primary outcome	16 & Figure 1
	13b	For each group, losses and exclusions after randomisation, together with reasons	Figure 1
Recruitment	14a	Dates defining the periods of recruitment and follow-up	16
	14b	Why the trial ended or was stopped	n/a
Baseline data	15	A table showing baseline demographic and clinical characteristics for each group	Table 1
Numbers analysed	16	For each group, number of participants (denominator) included in each analysis and whether the analysis was by original assigned groups	Table 2 and Figure
Outcomes and estimation	17a	For each primary and secondary outcome, results for each group, and the estimated effect size and its precision (such as 95% confidence interval)	Table 2
	17b	For binary outcomes, presentation of both absolute and relative effect sizes is recommended	n/a
Ancillary analyses	18	Results of any other analyses performed, including subgroup analyses and adjusted analyses, distinguishing pre-specified from exploratory	22-25
Harms	19	All important harms or unintended effects in each group (for specific guidance see CONSORT for harms)	25-26 & suppl Table
Discussion			
Limitations	20	Trial limitations, addressing sources of potential bias, imprecision, and, if relevant, multiplicity of analyses	27-8
Generalisability	21	Generalisability (external validity, applicability) of the trial findings	27-8
Interpretation	22	Interpretation consistent with results, balancing benefits and harms, and considering other relevant evidence	27-8
Other information			
Registration	23	Registration number and name of trial registry	6
Protocol	24	Where the full trial protocol can be accessed, if available	Reference 22
Funding	25	Sources of funding and other support (such as supply of drugs), role of funders	15 & 30

Online supplementary Tables

ONLINE SUPPLEMENTARY TABLE 1: BASELINE DEMOGRAPHIC & CLINICAL CHARACTERISTICS IN CONTROL & INTERVENTION ARMS

	Control (n=389)		Intervention (n=391)	
Gender				
Female	258	66.32%	266	68.03%
Male	129	33.16%	124	31.71%
Prefer not to say	1	0.26%	0	0.00%
Prefer to self-describe	1	0.26%	1	0.26%
Age (in years)	48.43	(14.34)	48.59	(14.45)
Body mass index (BMI)	27.15	(6.45)	26.83	(5.97)
BMI category				
Underweight	13	3.45%	13	3.42%
Healthy weight	142	37.67%	149	39.21%
Overweight or obese	222	58.89%	218	57.37%
Self-reported smoking status				
Never smoked	205	52.70%	209	53.73%
Ex-smoker	158	40.62%	158	40.62%
Current smoker	26	6.68%	22	5.66%
Alcohol consumption (units/week)				
None	188	48.33%	188	48.45%
1-14	183	47.04%	176	45.36%
15+	18	4.63%	24	6.19%
Ethnicity				
White	373	95.89%	371	94.88%
Mixed	6	1.54%	5	1.28%
Asian	7	1.80%	11	2.81%
Black	0	0.00%	1	0.26%
Other	2	0.51%	3	0.77%
Prefer not to say	1	0.26%	0	0.00%
Main work activity				
Employed full-time	158	40.62%	147	37.69%
Employed part-time	56	14.40%	53	13.59%
Student	13	3.34%	11	2.82%
Retired	70	17.99%	81	20.77%
Unemployed	8	2.06%	7	1.79%
Self-employed	29	7.46%	43	11.03%
Homemaker	17	4.37%	17	4.36%
Unemployed due to illness/disability	38	9.77%	31	7.95%
Highest level of education				
No formal education	4	1.03%	5	1.29%
Secondary school	61	15.72%	65	16.71%
Sixth form	24	6.19%	31	7.97%
Further education	113	29.12%	94	24.16%
Higher education	186	47.94%	194	49.87%
Relationship status				
Married/civil partnership	224	57.73%	203	52.19%
Living with partner	50	12.89%	76	19.54%
Widowed	10	2.58%	7	1.80%
Divorced/separated	24	6.19%	25	6.43%
Single	59	15.21%	55	14.14%
With partner, not living together	21	5.41%	23	5.91%
IBD diagnosis				
Crohn's disease	215	55.27%	217	55.50%
Other IBD	174	44.73%	174	44.50%
Rome IV criteria				
No	207	53.21%	200	51.28%
Yes	182	46.79%	190	48.72%
IBD operation history				
No	251	65.36%	245	64.47%
Yes	133	34.64%	135	35.53%
Stoma status				
No	363	93.32%	369	94.37%
Yes	26	6.68%	22	5.63%

Pouch status				
No	379	97.68%	375	96.40%
Yes	9	2.32%	14	3.60%
Fistula status				
No	329	84.58%	335	85.90%
Yes	24	6.17%	17	4.36%
Unsure	36	9.25%	38	9.74%
Biologic medications				
No	330	89.67%	325	87.84%
Yes	38	10.33%	45	12.16%
IBD medications				
No	318	81.75%	322	82.78%
Yes	71	18.25%	67	17.22%
Mental health conditions				
No	269	69.15%	272	69.57%
Yes	120	30.85%	119	30.43%
Physical health conditions				
No	223	57.33%	223	57.03%
Yes	166	42.67%	168	42.97%
Current pregnancy status				
No	353	99.16%	358	98.90%
Yes	3	0.84%	4	1.10%

Mean & standard deviation (SD) for continuous variables indicated by parentheses around SD; Absolute frequencies & column-wise percentages (not including missing values) for categorical data indicated by '%' for percentage summary data

ONLINE SUPPLEMENTARY TABLE 2: BASELINE OUTCOME VARIABLES IN CONTROL & INTERVENTION ARMS

	Control (n=389)		Intervention (n=391)	
UK-IBDQ score	63.76	(14.40)	64.80	(15.48)
Average pain intensity*	2.62	(2.20)	2.57	(2.15)
Lowest pain intensity*	1.31	(1.83)	1.24	(1.71)
Highest pain intensity*	4.30	(2.89)	4.31	(2.81)
Current pain intensity*	2.25	(2.35)	2.24	(2.32)
Faecal incontinence score**	9.30	(4.87)	8.88	(5.19)
IBD fatigue score	8.71	(3.55)	8.93	(3.49)
IBD control score	8.88	(4.16)	8.79	(4.48)
IBD control VAS score	6.58	(2.10)	6.47	(2.19)
EQ5D utility score	0.73	(0.22)	0.73	(0.22)
Illness perceptions	42.37	(11.38)	42.91	(12.12)
Self-efficacy for managing chronic disease	5.90	(2.11)	5.72	(1.97)
All-or-nothing thinking	13.21	(5.06)	13.53	(4.84)
Avoidance/resting behaviour	18.39	(5.61)	18.78	(5.89)
Patient Health Questionnaire-9 (for depression)	9.26	(5.68)	9.32	(5.66)
Visceral Sensitivity Index	39.31	(16.37)	38.96	(17.24)
IBD status (Calprotectin & IBD control score)				
Remission	49	16.07%	70	23.81%
Active	256	83.93%	224	76.19%
IBD status (Calprotectin only)				
Remission	242	79.34%	243	82.65%
Active	63	20.66%	51	17.35%
Depression status				
Not depressed	221	56.81%	219	56.01%
Depressed	168	43.19%	172	43.99%
Rome IV criteria				
No	207	53.21%	200	51.28%
Yes	182	46.79%	190	48.72%
Mobility (EQ5D)				
No problems	258	66.32%	263	67.26%
Slight problems	70	17.99%	75	19.18%
Moderate problems	42	10.80%	34	8.70%
Severe problems	17	4.37%	19	4.86%
Unable	2	0.51%	0	0.00%
Self-care (EQ5D)				
No problems	326	83.80%	328	83.89%
Slight problems	36	9.25%	39	9.97%
Moderate problems	21	5.40%	19	4.86%
Severe problems	6	1.54%	5	1.28%
Usual activities (EQ5D)				
No problems	182	46.79%	177	45.27%
Slight problems	123	31.62%	120	30.69%
Moderate problems	65	16.71%	65	16.62%
Severe problems	17	4.37%	22	5.63%
Unable	2	0.51%	7	1.79%
Pain/discomfort (EQ5D)				
None	115	29.56%	112	28.64%
Slight	167	42.93%	172	43.99%
Moderate	87	22.37%	87	22.25%
Severe	17	4.37%	13	3.32%
Extreme	3	0.77%	7	1.79%
Anxiety/depression (EQ5D)				
None	168	43.19%	173	44.25%
Slight	127	32.65%	119	30.43%
Moderate	76	19.54%	77	19.69%
Severe	9	2.31%	14	3.58%
Extreme	9	2.31%	8	2.05%

Mean & standard deviation (SD) for continuous variables indicated by parentheses around SD; Absolute frequencies & column-wise percentages for categorical data indicated by '%' for percentage summary data

*in the last seven days

**excludes 48 participants with stoma (thus, control arm [n=363] & intervention arm [n=369])

Online Supplementary Table 3: Primary & secondary trial outcomes at twelve months post-randomisation

	Control	(n= 389)			Intervention	(n= 391)						
	Included in analysis		Unadjusted mean & SD		Included in analysis		Unadjusted mean & SD		Effect estimate*	95%CI		P-value
Outcome	n	(%)	Mean	(SD)	n	%	Mean	(SD)				
UK-IBDQ	254	65.30	60.46	13.75	210	53.71	59.62	16.81	-1.62	-4.27	1.03	0.23
Global rating of symptom relief	250	64.27	3.98	2.85	207	52.94	4.24	2.81	0.30	-0.73	1.33	0.57
Average pain intensity**	252	64.78	2.46	2.04	209	53.45	2.39	2.24	-0.06	-0.68	0.57	0.86
Faecal incontinence score	232	59.64	8.54	4.93	195	49.87	7.13	5.26	-0.95	-1.60	-0.30	0.0009
IBD fatigue score	250	64.27	8.37	3.56	207	52.94	8.27	3.65	-0.19	-0.66	0.29	0.44
IBD control score	250	64.27	9.48	4.20	206	52.69	9.73	4.74	0.28	-0.89	1.44	0.64
IBD control VAS score	250	64.27	6.63	2.18	206	52.69	6.63	2.42	0.10	-0.25	0.45	0.57
EQ5D utility score	252	64.78	0.74	0.21	209	53.45	0.75	0.21	0.01	-0.02	0.04	0.49
Global rating of satisfaction	255	65.55	6.67	2.86	216	55.24	6.78	2.73	0.22	-0.84	1.27	0.69

SD, standard deviation; 95%CI, 95% confidence interval
*adjusted treatment effect estimate; **in the last seven days

Online supplementary Table 4: Subgroup analysis investigating the influence of baseline characteristics on the effect of the trial intervention on each primary outcome six months post-randomisation

Subgroup	Number included in analysis				Unadjusted, group-level, summary data				Effect estimate*	95%CI	P-value
	Control (n=389)		Intervention (n=391)		Control (n=389)		Intervention (n=391)				
	n	%	n	%	Mean	SD	Mean	SD			
<u>**UK-IBDQ**</u>											
IBD status											
In remission	45	11.57	61	15.60	49.76	10.16	48.54	9.82	-2.88	(-6.84, 1.07)	0.56
Active	251	64.52	192	49.10	63.47	13.77	63.42	15.81	-1.57	(-3.54, 0.40)	
PHQ-9											
Not depressed	200	51.41	176	45.01	56.94	11.94	54.39	12.54	-2.56	(-4.65, -0.47)	0.18
Depressed	158	40.62	129	32.99	68.61	14.67	69.66	16.22	-0.37	(-2.77, 2.03)	
VSI	358	92.03	305	78.01	62.09	14.42	60.85	16.08	-0.01	(-0.11, 0.08)	0.76
IBS-like symptoms (Rome IV criteria)											
No	191	49.10	162	41.43	56.37	12.19	56.33	15.19	-0.41	(-2.56, 1.74)	0.09
Yes	167	42.93	142	36.32	68.62	14.02	65.94	15.61	-3.13	(-5.44, -0.81)	
<u>**Global Rating of Symptom Relief**</u>											
IBD status											
In remission	44	11.31	61	15.60	3.66	3.09	4.74	2.90	1.19	(0.15, 2.24)	0.23
Active	250	64.27	192	49.10	3.68	2.69	4.15	2.74	0.47	(-0.03, 0.98)	
PHQ-9											
Not depressed	196	50.39	176	45.01	3.66	2.77	4.34	2.86	0.74	(0.18, 1.30)	0.18
Depressed	158	40.62	129	32.99	3.65	2.73	3.84	2.72	0.16	(-0.47, 0.80)	
VSI	354	91.00	305	78.01	3.65	2.75	4.13	2.81	-0.01	(-0.03, 0.02)	0.62
IBS-like symptoms (Rome IV criteria)											
No	188	48.33	162	41.43	3.93	2.92	4.27	2.75	0.28	(-0.29, 0.85)	0.29
Yes	166	42.67	142	36.32	3.34	2.52	3.96	2.89	0.74	(0.12, 1.35)	

SD, standard deviation; 95%CI, 95% confidence interval; IBD, inflammatory bowel disease; PHQ-9, patient health questionnaire-9; VSI, visceral sensitivity index; IBS, irritable bowel syndrome
*adjusted treatment effect estimate

Online Supplementary Table 5: UK-IBDQ

Missing data management method	Delta	Effect estimate*	95%CI		P-value
Six months post-randomisation					
POAM	-	-1.67	-4.13	0.80	0.19
MAR	-	-1.42	-3.51	0.67	0.18
Missing values worse than observed values in intervention arm and control arm					
MNAR (25%)	0.70	-1.34	-3.48	0.79	0.22
MNAR (50%)	1.39	-1.25	-3.43	0.92	0.26
MNAR (75%)	2.09	-1.16	-3.38	1.06	0.31
MNAR (100%)	2.79	-1.05	-3.33	1.23	0.37
Missing values worse than observed values in intervention arm but better than observed values in control arm					
MNAR (25%)	0.70	-1.19	-3.34	0.96	0.28
MNAR (50%)	1.39	-0.97	-3.18	1.23	0.39
MNAR (75%)	2.09	-0.75	-3.02	1.51	0.51
MNAR (100%)	2.79	-0.54	-2.87	1.79	0.65
Twelve months post-randomisation					
POAM	-	-1.62	-4.27	1.03	0.23
MAR	-	-1.37	-3.72	0.99	0.26
Missing values worse than observed values in intervention arm and control arm					
MNAR (25%)	0.60	-1.32	-3.72	1.08	0.28
MNAR (50%)	1.19	-1.25	-3.69	1.19	0.32
MNAR (75%)	1.79	-1.18	-3.66	1.30	0.35
MNAR (100%)	2.39	-1.09	-3.63	1.44	0.40
Missing values worse than observed values in intervention arm but better than observed values in control arm					
MNAR (25%)	0.60	-0.85	-3.25	1.55	0.49
MNAR (50%)	1.19	-0.35	-2.80	2.10	0.78
MNAR (75%)	1.79	0.15	-2.35	2.65	0.91
MNAR (100%)	2.39	0.63	-1.95	3.20	0.63

95%CI, 95% confidence interval; POAM, primary outcome analysis model; MAR, missing at random; MNAR, missing not at random
*adjusted treatment effect estimate

Online Supplementary Table 6: global rating of symptom relief

Missing data management method	Delta	Effect estimate*	95%CI		P-value
Six months post-randomisation					
POAM	-	0.44	-0.56	1.44	0.39
MAR	-	0.46	-0.20	1.12	0.17
Missing values worse than observed values in intervention arm and control arm					
MNAR (25%)	0.04	0.46	-0.21	1.12	0.18
MNAR (50%)	0.09	0.45	-0.22	1.12	0.19
MNAR (75%)	0.13	0.45	-0.23	1.12	0.19
MNAR (100%)	0.18	0.44	-0.24	1.11	0.21
Missing values worse than observed values in intervention arm but better than observed values in control arm					
MNAR (25%)	0.04	0.45	-0.22	1.11	0.19
MNAR (50%)	0.09	0.43	-0.24	1.10	0.20
MNAR (75%)	0.13	0.42	-0.26	1.09	0.22
MNAR (100%)	0.18	0.41	-0.27	1.08	0.24
Twelve months post-randomisation					
POAM	-	0.30	-0.73	1.33	0.57
MAR	-	0.32	-0.40	1.03	0.39
Missing values worse than observed values in intervention arm and control arm					
MNAR (25%)	0.04	0.31	-0.41	1.03	0.40
MNAR (50%)	0.09	0.31	-0.42	1.03	0.41
MNAR (75%)	0.13	0.30	-0.43	1.03	0.42
MNAR (100%)	0.18	0.29	-0.44	1.03	0.43
Missing values worse than observed values in intervention arm but better than observed values in control arm					
MNAR (25%)	0.04	0.28	-0.44	1.00	0.45
MNAR (50%)	0.09	0.24	-0.48	0.96	0.51
MNAR (75%)	0.13	0.20	-0.52	0.93	0.58
MNAR (100%)	0.18	0.17	-0.57	0.90	0.65

95%CI, 95% confidence interval; POAM, primary outcome analysis model; MAR, missing at random; MNAR, missing not at random
*adjusted treatment effect estimate

Online Supplementary Table 7: Sensitivity analysis of primary & secondary trial outcomes including participants with post-randomisation assessments after the six-month follow-up window of 8 weeks (plus a two-week 'grace period')

	Control		**(n= 389)**		**Intervention**		**(n= 391)**		
	Included in analysis		Unadjusted mean (SD)		Included in analysis		Unadjusted mean (SD)		
Outcome	n	(%)	Mean	(SD)	n	%	Mean	(SD)	Ef
UK-IBDQ									
Excluding (ie, POAM)	358	92.03	62.09	14.42	305	78.01	60.85	16.08	
Including	359	92.29	62.10	14.40	306	78.26	60.87	16.05	
Global rating of symptom relief									
Excluding (ie, POAM)	354	91.00	3.65	2.75	305	78.01	4.13	2.81	
Including	356	91.52	3.66	2.74	306	78.26	4.11	2.82	
Average pain intensity**									
Excluding (ie, POAM)	356	91.52	2.54	2.23	305	78.01	2.37	2.15	
Including	358	92.03	2.55	2.23	306	78.26	2.37	2.15	
Faecal incontinence score									
Excluding (ie, POAM)	332	85.35	8.81	5.05	287	73.40	7.74	5.22	
Including	334	85.86	8.82	5.03	288	73.66	7.74	5.21	
IBD fatigue score									
Excluding (ie, POAM)	355	91.26	8.57	3.57	305	78.01	8.28	3.83	
Including	357	91.77	8.58	3.56	306	78.26	8.28	3.83	
IBD control score									
Excluding (ie, POAM)	354	91.00	8.92	4.42	305	78.01	9.79	4.68	
Including	356	91.52	8.91	4.42	306	78.26	9.78	4.68	
IBD control VAS score									
Excluding (ie, POAM)	354	91.00	6.59	2.17	305	78.01	6.58	2.25	
Including	356	91.52	6.60	2.17	306	78.26	6.57	2.24	
EQ5D utility score									
Excluding (ie, POAM)	347	89.20	0.71	0.23	298	76.21	0.75	0.21	
Including	349	89.72	0.71	0.23	298	76.21	0.75	0.21	

SD, standard deviation; POAM, primary outcome analysis model excluding participants with follow-up assessments more than 10 weeks after six months post-randomisation; VAS, visual analogue scale
*adjusted treatment effect estimate; **in the last seven days

Online Supplementary Table 8: Sensitivity analysis of primary & secondary trial outcomes including participants with post-randomisation assessments after the 12-month follow-up window of 4 weeks (plus a two-week 'grace period')

	Control		(n= 389)		Intervention		(n= 391)					
	Included in analysis		Unadjusted mean (SD)		Included in analysis		Unadjusted mean (SD)					
Outcome	n	(%)	Mean	(SD)	n	%	Mean	(SD)	Effect estimate*	95%CI		P-value
UK-IBDQ												
Excluding (ie, POAM)	254	65.30	60.46	13.75	210	53.71	59.62	16.81	-1.62	-4.27	1.03	0.23
Including	259	66.58	60.44	13.75	219	56.01	59.46	16.54	-1.91	-4.82	1.01	0.20
Global rating of symptom relief												
Excluding (ie, POAM)	250	64.27	3.98	2.85	207	52.94	4.24	2.81	0.30	-0.73	1.33	0.57
Including	255	65.55	3.96	2.87	216	55.24	4.24	2.82	0.32	-0.74	1.38	0.55
Average pain intensity**												
Excluding (ie, POAM)	252	64.78	2.46	2.04	209	53.45	2.39	2.24	-0.06	-0.68	0.57	0.86
Including	257	66.07	2.44	2.05	218	55.75	2.38	2.21	-0.07	-0.63	0.49	0.80
Faecal incontinence score												
Excluding (ie, POAM)	232	59.64	8.54	4.93	195	49.87	7.13	5.26	-0.95	-1.60	-0.30	0.004
Including	237	60.93	8.50	4.91	204	52.17	7.12	5.19	-1.01	-1.65	-0.38	0.002
IBD fatigue score												
Excluding (ie, POAM)	250	64.27	8.37	3.56	207	52.94	8.27	3.65	-0.19	-0.66	0.29	0.44
Including	255	65.55	8.34	3.55	216	55.24	8.29	3.63	-0.16	-0.63	0.31	0.50
IBD control score												
Excluding (ie, POAM)	250	64.27	9.48	4.20	206	52.69	9.73	4.74	0.28	-0.89	1.44	0.64
Including	255	65.55	9.48	4.17	215	54.99	9.81	4.69	0.39	-0.88	1.65	0.55
IBD control VAS score												
Excluding (ie, POAM)	250	64.27	6.63	2.18	206	52.69	6.63	2.42	0.10	-0.25	0.45	0.57
Including	255	65.55	6.65	2.18	215	54.99	6.68	2.40	0.14	-0.21	0.48	0.44
EQ5D utility score												
Excluding (ie, POAM)	246	63.24	0.74	0.21	201	51.41	0.76	0.21	0.01	-0.02	0.04	0.48
Including	252	64.78	0.74	0.21	209	53.45	0.75	0.21	0.01	-0.02	0.04	0.49

SD, standard deviation; POAM, primary outcome analysis model excluding participants with follow-up assessments more than 10 weeks after six months post-randomisation; VAS, visual analogue scale
*adjusted treatment effect estimate; **in the last seven days

Online supplementary Table 9: Sensitivity subgroup analysis investigating the influence of re-defined baseline characteristics on the effect of the trial intervention on each primary outcome six months post-randomisation

Subgroup	Number included in analysis				Unadjusted, group-level, summary data				Effect estimate*	95%CI	P-value
	Control (n=389)		Intervention (n=391)		Control (n=389)		Intervention (n=391)				
	n	%	n	%	Mean	SD	Mean	SD			
<u>UK-IBDQ</u>											
IBD status (faecal calprotectin only)**											
In remission	234	60.15	211	53.96	60.91	13.45	59.75	15.65	-1.77	(-3.70, 0.17)	0.61
Active	62	15.94	42	10.74	63.15	16.53	60.21	17.36	-2.95	(-7.06, 1.15)	
IBS-like symptoms (Rome IV criteria)***											
No	127	32.65	109	27.88	55.49	10.93	56.37	15.61	-0.03	(-2.55, 2.48)	0.044
Yes	107	27.51	101	25.83	67.36	13.37	63.31	14.99	-3.82	(-6.49, -1.15)	
<u>Global Rating of Symptom Relief</u>											
IBD status (faecal calprotectin only)**											
In remission	232	59.64	211	53.96	3.59	2.68	4.29	2.76	0.73	(0.22, 1.23)	0.40
Active	62	15.94	42	10.74	4.03	2.98	4.31	2.94	0.23	(-0.83, 1.28)	
IBS-like symptoms (Rome IV criteria)***											
No	125	32.13	109	27.88	3.72	2.86	4.21	2.73	0.50	(-0.18, 1.18)	0.27
Yes	107	27.51	101	25.83	3.43	2.47	4.37	2.82	1.07	(0.34, 1.79)	

SD, standard deviation; 95%CI, 95% confidence interval; IBD, inflammatory bowel disease; PHQ-9, patient health questionnaire-9; VSI, visceral sensitivity index; IBS, irritable bowel syndrome

*adjusted treatment effect estimate; **active IBD defined as faecal calprotectin score ≥20ug/g; ***IBS-like symptoms (as classified using ROME IV criteria for participants in remission only)

Supplementary Table 10: Frequency (n) and percentage (%) of adverse events (AE) and serious adverse events considered related/unrelated to the trial, stratified by trial arm

	Control (n=389)		Intervention (n=391)	
	n	%	n	%
AEs				
Total number of AEs	337	86.63	351	89.77
Number of participants experiencing at least one AE	193	49.61	188	48.08
SAEs				
Total number of SAEs	79	20.31	55	14.07
Mild	0	0.00	0	0.00
Moderate	0	0.00	0	0.00
Severe	73	18.77	45	11.51
Life-threatening	6	1.54	10	2.56
Number of participants experiencing at least one SAE	68	17.48	48	12.28
Possibly related to the trial	0	0.00	1	0.26
Not related to the trial	68	17.48	47	12.02

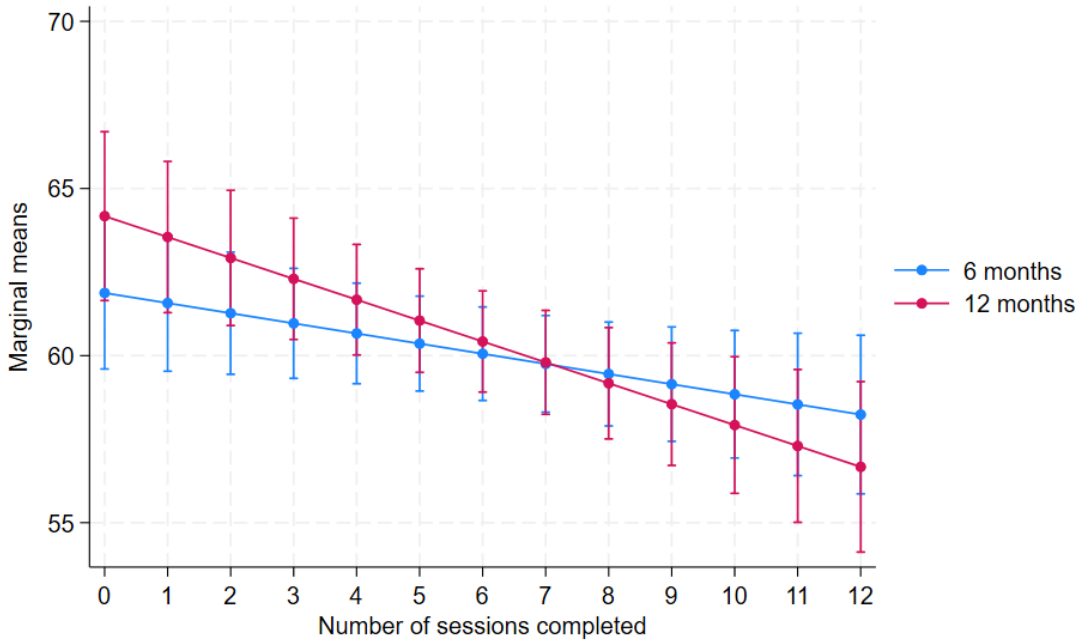
AEs, adverse events; SAEs, serious adverse events

SUPPLEMENTARY TABLE 11. DOSE-RESPONSE RELATIONSHIP BETWEEN NUMBER OF INTERVENTION SESSIONS COMPLETED AND IBD-RELATED QUALITY OF LIFE AMONGST INTERVENTION ARM PARTICIPANTS SIX- & 12- MONTHS POST-RANDOMISATION

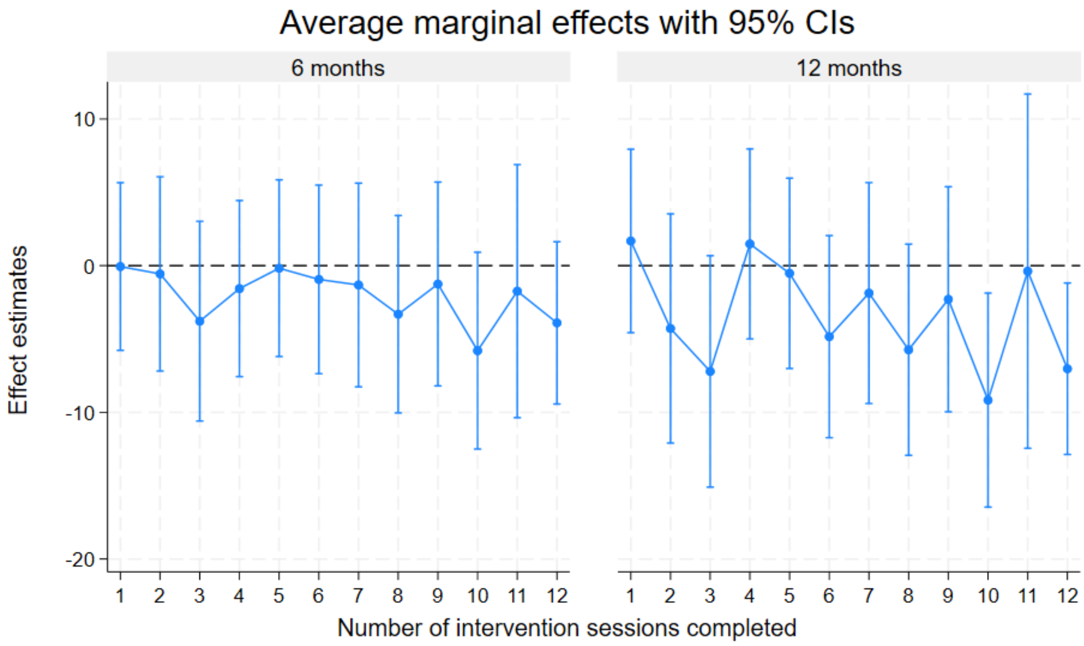
Sessions	Coefficient	Std Error	95%CI		P-value
Continuous*					
Six months	-0.32	0.16	-0.63	-0.003	0.048
12 months	-0.63	0.17	-0.98	-0.29	0.0003
Categorical**					
Six months					
1 session	-0.05	2.92	-5.77	5.66	0.99
2 sessions	-0.56	3.38	-7.18	6.06	0.87
3 sessions	-3.78	3.47	-10.59	3.03	0.28
4 sessions	-1.56	3.06	-7.56	4.44	0.61
5 sessions	-0.17	3.07	-6.19	5.85	0.96
6 sessions	-0.93	3.28	-7.36	5.49	0.78
7 sessions	-1.32	3.54	-8.26	5.63	0.71
8 sessions	-3.31	3.43	-10.04	3.42	0.34
9 sessions	-1.25	3.54	-8.20	5.70	0.72
10 sessions	-5.79	3.42	-12.50	0.92	0.09
11 sessions	-1.74	4.40	-10.36	6.89	0.69
12 sessions	-3.90	2.82	-9.43	1.63	0.17
12 months					
1 session	1.69	3.19	-4.57	7.94	0.60
2 sessions	-4.28	3.99	-12.09	3.53	0.28
3 sessions	-7.21	4.03	-15.10	0.68	0.07
4 sessions	1.49	3.30	-4.99	7.96	0.65
5 sessions	-0.52	3.31	-7.01	5.96	0.87
6 sessions	-4.84	3.52	-11.73	2.05	0.17
7 sessions	-1.87	3.84	-9.40	5.66	0.63
8 sessions	-5.73	3.67	-12.93	1.47	0.12
9 sessions	-2.29	3.92	-9.96	5.38	0.56
10 sessions	-9.17	3.72	-16.46	-1.87	0.01
11 sessions	-0.37	6.16	-12.45	11.70	0.95
12 sessions	-7.03	2.98	-12.87	-1.18	0.02

Footnote: Std Error, standard error; 95%CI, 95% confidence intervals; * as continuous variable (0-12); ** as categorical variable (0-12)

SUPPLEMENTRY FIGURE 1A. DOSE-RESPONSE RELATIONSHIP BETWEEN NUMBER OF INTERVENTION SESSIONS COMPLETED (AS A CONTINUOUS VARIABLE) & IBD-RELATED QUALITY OF LIFE AMONGST INTERVENTION PARTICIPANTS SIX- & 12-MONTHS POST-RANDOMISATION



SUPPLEMENTRY FIGURE 1B. DOSE-RESPONSE RELATIONSHIP BETWEEN NUMBER OF INTERVENTION SESSIONS COMPLETED (AS A CATEGORICAL VARIABLE) & IBD-RELATED QUALITY OF LIFE AMONGST INTERVENTION PARTICIPANTS SIX- & 12-MONTHS POST-RANDOMISATION



Reference level: 0 sessions

SUPPLEMENTRY TABLE 12A. EFFECT OF THE IBD-BOOST INTERVENTION ON THE VAIZEY INCONTINENCE SCORE AT EACH LEVEL OF THE VAIZEY INCONTINENCE SCORE AT BASELINE FOR THE ENTIRE TRIAL SAMPLE SIX- & 12-MONTHS POST-RANDOMISATION

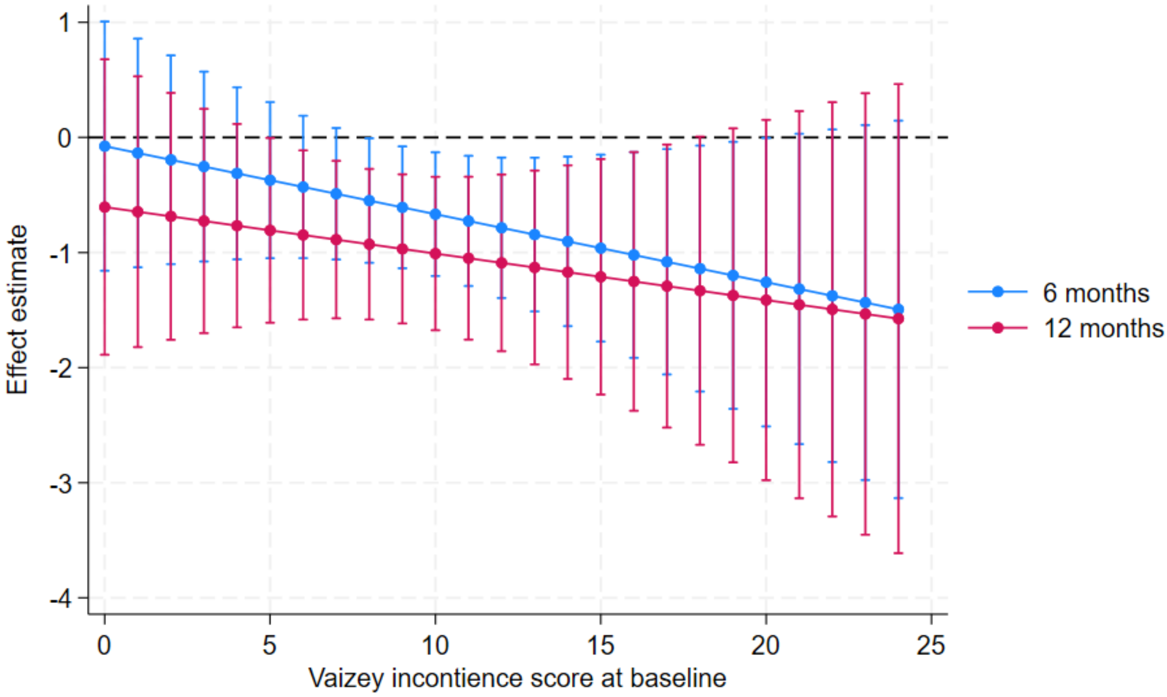
Moderator value	Coefficient	Std Error	95%CI		P-value
Six months					
0	-0.08	0.55	-1.16	1.01	0.89
1	-0.14	0.51	-1.13	0.86	0.79
2	-0.19	0.46	-1.10	0.71	0.68
3	-0.25	0.42	-1.08	0.57	0.55
4	-0.31	0.38	-1.06	0.44	0.41
5	-0.37	0.35	-1.05	0.31	0.28
6	-0.43	0.32	-1.05	0.19	0.17
7	-0.49	0.29	-1.06	0.08	0.09
8	-0.55	0.28	-1.09	-0.01	0.05
9	-0.61	0.27	-1.14	-0.08	0.02
10	-0.67	0.27	-1.20	-0.13	0.01
11	-0.73	0.29	-1.29	-0.16	0.01
12	-0.79	0.31	-1.40	-0.17	0.01
13	-0.84	0.34	-1.51	-0.18	0.01
14	-0.90	0.38	-1.64	-0.17	0.02
15	-0.96	0.41	-1.77	-0.15	0.02
16	-1.02	0.46	-1.91	-0.13	0.03
17	-1.08	0.50	-2.06	-0.10	0.03
18	-1.14	0.55	-2.21	-0.07	0.04
19	-1.20	0.59	-2.36	-0.04	0.04
20	-1.26	0.64	-2.51	-0.00	0.05
21	-1.32	0.69	-2.67	0.03	0.06
22	-1.38	0.74	-2.82	0.07	0.06
23	-1.44	0.79	-2.98	0.11	0.07
24	-1.49	0.84	-3.13	0.15	0.07
12 months					
0	-0.60	0.66	-1.89	0.68	0.36
1	-0.65	0.60	-1.82	0.53	0.28
2	-0.69	0.55	-1.76	0.39	0.21
3	-0.73	0.50	-1.70	0.25	0.14
4	-0.77	0.45	-1.65	0.12	0.09
5	-0.81	0.41	-1.61	-0.00	0.05
6	-0.85	0.38	-1.58	-0.11	0.02
7	-0.89	0.35	-1.57	-0.20	0.01
8	-0.93	0.33	-1.58	-0.27	0.005
9	-0.97	0.33	-1.62	-0.32	0.003
10	-1.01	0.34	-1.68	-0.34	0.003
11	-1.05	0.36	-1.76	-0.34	0.004
12	-1.09	0.39	-1.86	-0.32	0.005
13	-1.13	0.43	-1.97	-0.29	0.009
14	-1.17	0.47	-2.10	-0.24	0.01
15	-1.21	0.52	-2.23	-0.19	0.02
16	-1.25	0.57	-2.38	-0.13	0.03
17	-1.29	0.63	-2.52	-0.06	0.04
18	-1.33	0.68	-2.67	0.01	0.05
19	-1.37	0.74	-2.82	0.08	0.06
20	-1.41	0.80	-2.98	0.15	0.08
21	-1.45	0.86	-3.13	0.23	0.09
22	-1.49	0.92	-3.29	0.31	0.10
23	-1.53	0.98	-3.45	0.38	0.12
24	-1.57	1.04	-3.61	0.46	0.13

Footnote: Std Error, standard error; 95%CI, 95% confidence intervals

Three-way interaction (i.e., trial arm-by-assessment-by-baseline moderator): coefficient = 0.019 (SE = 0.06), p=0.77

SUPPLEMENTRY FIGURE 2A. EFFECT OF THE IBD-BOOST INTERVENTION ON THE VAIZEY INCONTINENCE SCORE AT EACH LEVEL OF THE VAIZEY INCONTINENCE SCORE AT BASELINE FOR THE ENTIRE TRIAL SAMPLE SIX- & 12-

MONTHS POST-RANDOMISATION



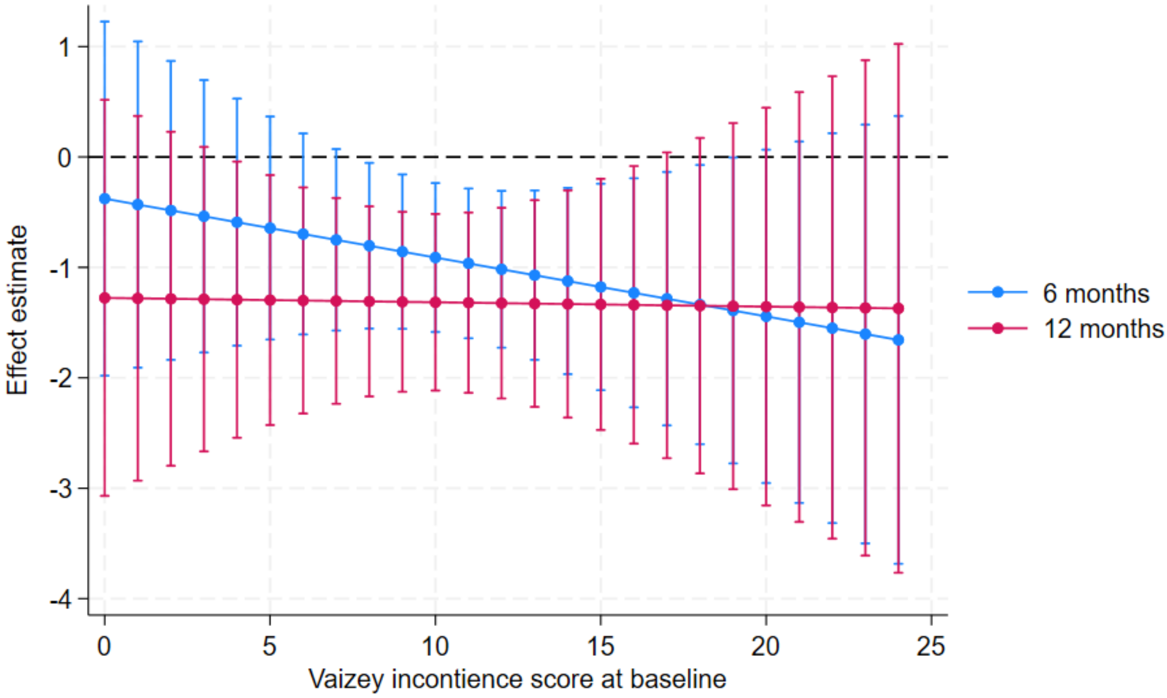
SUPPLEMENTRY TABLE 12B. EFFECT OF THE IBD-BOOST INTERVENTION ON THE VAIZEY INCONTINENCE SCORE AT EACH LEVEL OF THE VAIZEY INCONTINENCE SCORE AT BASELINE SIX- & 12-MONTHS POST-RANDOMISATION FOR TRIAL PARTICIPANTS WITH INCONTINENCE SYMPTOMS UPON ENROLMENT

Moderator value	Coefficient	Std Error	95%CI		P-value
Six months					
0	-0.377	0.82	-1.98	1.23	0.65
1	-0.431	0.75	-1.91	1.05	0.57
2	-0.484	0.69	-1.84	0.87	0.48
3	-0.537	0.63	-1.77	0.70	0.39
4	-0.590	0.57	-1.71	0.53	0.30
5	-0.644	0.52	-1.65	0.37	0.21
6	-0.697	0.46	-1.61	0.21	0.13
7	-0.750	0.42	-1.57	0.07	0.07
8	-0.804	0.38	-1.55	-0.05	0.04
9	-0.857	0.36	-1.56	-0.16	0.02
10	-0.910	0.34	-1.58	-0.24	0.008
11	-0.964	0.35	-1.64	-0.29	0.005
12	-1.017	0.36	-1.73	-0.31	0.005
13	-1.070	0.39	-1.84	-0.30	0.006
14	-1.124	0.43	-1.97	-0.28	0.009
15	-1.177	0.48	-2.11	-0.24	0.01
16	-1.230	0.53	-2.27	-0.19	0.02
17	-1.283	0.59	-2.43	-0.14	0.03
18	-1.337	0.65	-2.60	-0.07	0.04
19	-1.390	0.71	-2.77	-0.01	0.05
20	-1.443	0.77	-2.95	0.07	0.06
21	-1.497	0.83	-3.13	0.14	0.07
22	-1.550	0.90	-3.31	0.21	0.09
23	-1.603	0.97	-3.50	0.29	0.10
24	-1.657	1.03	-3.68	0.37	0.11
12 months					
0	-1.276	0.92	-3.07	0.52	0.16
1	-1.280	0.84	-2.93	0.37	0.13
2	-1.284	0.77	-2.80	0.23	0.10
3	-1.288	0.70	-2.67	0.09	0.07
4	-1.292	0.64	-2.54	-0.04	0.04
5	-1.296	0.58	-2.43	-0.16	0.03
6	-1.300	0.52	-2.32	-0.28	0.01
7	-1.304	0.47	-2.24	-0.37	0.006
8	-1.308	0.44	-2.17	-0.45	0.003
9	-1.312	0.42	-2.13	-0.50	0.002
10	-1.315	0.41	-2.12	-0.52	0.001
11	-1.319	0.42	-2.14	-0.50	0.002
12	-1.323	0.44	-2.19	-0.46	0.003
13	-1.327	0.48	-2.26	-0.39	0.005
14	-1.331	0.53	-2.36	-0.30	0.01
15	-1.335	0.58	-2.47	-0.20	0.02
16	-1.339	0.64	-2.60	-0.08	0.04
17	-1.343	0.71	-2.73	0.04	0.06
18	-1.347	0.78	-2.87	0.17	0.08
19	-1.351	0.85	-3.01	0.31	0.11
20	-1.355	0.92	-3.16	0.45	0.14
21	-1.359	0.99	-3.31	0.59	0.17
22	-1.363	1.07	-3.46	0.73	0.20
23	-1.367	1.15	-3.61	0.88	0.23
24	-1.371	1.22	-3.77	1.02	0.26

Footnote: Std Error, standard error; 95%CI, 95% confidence intervals

Three-way interaction (i.e., trial arm-by-assessment-by-baseline moderator): coefficient = 0.049 (SE = 0.08), p=0.55

SUPPLEMENTRY FIGURE 2B. EFFECT OF THE IBD-BOOST INTERVENTION ON THE VAIZEY INCONTINENCE SCORE AT EACH LEVEL OF THE VAIZEY INCONTINENCE SCORE AT BASELINE SIX- & 12-MONTHS POST-RANDOMISATION FOR TRIAL PARTICIPANTS WITH INCONTINENCE SYMPTOMS UPON ENROLMENT



SUPPLEMENTRY TABLE 13. BASELINE DEMOGRAPHIC & CLINICAL CHARACTERISTICS IN PARTICIPANTS WITH & WITHOUT OBSERVED UK-IBDQ DATA AT SIX MONTHS POST-RANDOMISATION

	UK-IBDQ Observed (N=663)				UK-IBDQ Missing (N=117)				Total (N=780)			
	Control (n=358)		Intervention (n=305)		Control (n=31)		Intervention (n=86)		Control (n=389)		Intervention (n=391)	
Gender												
Female	234	65.36%	207	67.87%	24	77.42%	59	68.60%	258	66.32%	266	68.03%
Male	122	34.08%	98	32.13%	7	22.58%	26	30.23%	129	33.16%	124	31.71%
Prefer not to say	1	0.28%	0	0.00%	0	0.00%	0	0.00%	1	0.26%	0	0.00%
Prefer to self-describe	1	0.28%	0	0.00%	0	0.00%	1	1.16%	1	0.26%	1	0.26%
Age (in years)	48.99	(14.06)	49.73	(14.50)	42.03	(16.15)	44.57	(13.60)	48.43	(14.34)	48.59	(14.45)
Body mass index (BMI)	27.24	(6.54)	26.54	(5.71)	26.09	(5.28)	27.86	(6.77)	27.15	(6.45)	26.83	(5.97)
BMI category												
Underweight	12	3.45%	11	3.70%	1	3.45%	2	2.41%	13	3.45%	13	3.42%
Healthy weight	128	36.78%	118	39.73%	14	48.28%	31	37.35%	142	37.67%	149	39.21%
Overweight	208	59.77%	168	56.57%	14	48.28%	50	60.24%	222	58.89%	218	57.37%
Self-reported smoking status												
Never smoked	187	52.23%	175	57.76%	18	58.06%	34	39.53%	205	52.70%	209	53.73%
Ex-smoker	147	41.06%	114	37.62%	11	35.48%	44	51.16%	158	40.62%	158	40.62%
Current smoker	24	6.70%	14	4.62%	2	6.45%	8	9.30%	26	6.68%	22	5.66%
Alcohol consumption (units/week)												
None	168	46.93%	147	48.68%	20	64.52%	41	47.67%	188	48.33%	188	48.45%
1-14	173	48.32%	134	44.37%	10	32.26%	42	48.84%	183	47.04%	176	45.36%
15+	17	4.75%	21	6.95%	1	3.23%	3	3.49%	18	4.63%	24	6.19%
Ethnicity												
White	343	95.81%	290	95.08%	30	96.77%	81	94.19%	373	95.89%	371	94.88%
Mixed	5	1.40%	4	1.31%	1	3.23%	1	1.16%	6	1.54%	5	1.28%
Asian	7	1.96%	9	2.95%	0	0.00%	2	2.33%	7	1.80%	11	2.81%
Black	0	0.00%	1	0.33%	0	0.00%	0	0.00%	0	0.00%	1	0.26%
Other	2	0.56%	1	0.33%	0	0.00%	2	2.33%	2	0.51%	3	0.77%
Prefer not to say	1	0.28%	0	0.00%	0	0.00%	0	0.00%	1	0.26%	0	0.00%
Main work activity												
Employed full-time	141	39.39%	106	34.87%	17	54.84%	41	47.67%	158	40.62%	147	37.69%
Employed part-time	52	14.53%	41	13.49%	4	12.90%	12	13.95%	56	14.40%	53	13.59%
Student	12	3.35%	10	3.29%	1	3.23%	1	1.16%	13	3.34%	11	2.82%
Retired	65	18.16%	73	24.01%	5	16.13%	8	9.30%	70	17.99%	81	20.77%
Unemployed	8	2.23%	7	2.30%	0	0.00%	0	0.00%	8	2.06%	7	1.79%
Self-employed	28	7.82%	31	10.20%	1	3.23%	12	13.95%	29	7.46%	43	11.03%
Homemaker	17	4.75%	11	3.62%	0	0.00%	6	6.98%	17	4.37%	17	4.36%
Unemployed due to illness/disability	35	9.78%	25	8.22%	3	9.68%	6	6.98%	38	9.77%	31	7.95%
Highest level of education												

No formal education	4	1.12%	3	0.99%	0	0.00%	2	2.33%	4	1.03%	5	1.29%
Secondary school	56	15.69%	45	14.85%	5	16.13%	20	23.26%	61	15.72%	65	16.71%
Sixth form	20	5.60%	25	8.25%	4	12.90%	6	6.98%	24	6.19%	31	7.97%
Further education	105	29.41%	70	23.10%	8	25.81%	24	27.91%	113	29.12%	94	24.16%
Higher education	172	48.18%	160	52.81%	14	45.16%	34	39.53%	186	47.94%	194	49.87%
Relationship status												
Married/civil partnership	213	59.66%	161	53.14%	11	35.48%	42	48.84%	224	57.73%	203	52.19%
Living with partner	43	12.04%	53	17.49%	7	22.58%	23	26.74%	50	12.89%	76	19.54%
Widowed	7	1.96%	7	2.31%	3	9.68%	0	0.00%	10	2.58%	7	1.80%
Divorced/separated	23	6.44%	21	6.93%	1	3.23%	4	4.65%	24	6.19%	25	6.43%
Single	54	15.13%	45	14.85%	5	16.13%	10	11.63%	59	15.21%	55	14.14%
With partner, not living together	17	4.76%	16	5.28%	4	12.90%	7	8.14%	21	5.41%	23	5.91%
IBD diagnosis												
Crohn's disease	200	55.87%	173	56.72%	15	48.39%	44	51.16%	215	55.27%	217	55.50%
Other IBD	158	44.13%	132	43.28%	16	51.61%	42	48.84%	174	44.73%	174	44.50%
Rome IV criteria												
No	191	53.35%	162	53.29%	16	51.61%	38	44.19%	207	53.21%	200	51.28%
Yes	167	46.65%	142	46.71%	15	48.39%	48	55.81%	182	46.79%	190	48.72%
IBD operation history												
No	226	63.84%	187	63.39%	25	83.33%	58	68.24%	251	65.36%	245	64.47%
Yes	128	36.16%	108	36.61%	5	16.67%	27	31.76%	133	34.64%	135	35.53%
Stoma status												
No	334	93.30%	287	94.10%	29	93.55%	82	95.35%	363	93.32%	369	94.37%
Yes	24	6.70%	18	5.90%	2	6.45%	4	4.65%	26	6.68%	22	5.63%
Pouch status												
No	349	97.76%	290	95.71%	30	96.77%	85	98.84%	379	97.68%	375	96.40%
Yes	8	2.24%	13	4.29%	1	3.23%	1	1.16%	9	2.32%	14	3.60%
Fistula status												
No	302	84.36%	263	86.51%	27	87.10%	72	83.72%	329	84.58%	335	85.90%
Yes	23	6.42%	15	4.93%	1	3.23%	2	2.33%	24	6.17%	17	4.36%
Unsure	33	9.22%	26	8.55%	3	9.68%	12	13.95%	36	9.25%	38	9.74%
Biologic medications												
No	304	89.68%	254	86.99%	26	89.66%	71	91.03%	330	89.67%	325	87.84%
Yes	35	10.32%	38	13.01%	3	10.34%	7	8.97%	38	10.33%	45	12.16%
IBD medications												
No	291	81.28%	253	83.50%	27	87.10%	69	80.23%	318	81.75%	322	82.78%
Yes	67	18.72%	50	16.50%	4	12.90%	17	19.77%	71	18.25%	67	17.22%
Mental health conditions												
No	251	70.11%	215	70.49%	18	58.06%	57	66.28%	269	69.15%	272	69.57%
Yes	107	29.89%	90	29.51%	13	41.94%	29	33.72%	120	30.85%	119	30.43%

Physical health conditions												
No	202	56.42%	173	56.72%	21	67.74%	50	58.14%	223	57.33%	223	57.03%
Yes	156	43.58%	132	43.28%	10	32.26%	36	41.86%	166	42.67%	168	42.97%
Current pregnancy status												
No	324	99.39%	275	98.57%	29	96.67%	83	100.00%	353	99.16%	358	98.90%
Yes	2	0.61%	4	1.43%	1	3.33%	0	0.00%	3	0.84%	4	1.10%

Mean & standard deviation (SD) for continuous variables indicated by parentheses around SD; Absolute frequencies & column-wise percentages for categorical data indicated by '%' for percentage summary data

SUPPLEMENTRY TABLE 14. BASELINE OUTCOME VARIABLES IN TRIAL PARTICIPANTS WITH & WITHOUT OBSERVED UK-IBDQ DATA AT SIX MONTHS POST-RANDOMISATION

	UK-IBDQ Observed (N=663)				UK-IBDQ Missing (N=117)				Total (N=780)			
	Control (n=358)		Intervention (n=305)		Control (n=31)		Intervention (n=86)		Control (n=389)		Intervention (n=391)	
UK-IBDQ score	63.96	(14.48)	64.83	(15.87)	61.45	(13.36)	64.72	(14.14)	63.76	(14.40)	64.80	(15.48)
Average pain intensity*	2.63	(2.22)	2.50	(2.15)	2.48	(2.00)	2.84	(2.15)	2.62	(2.20)	2.57	(2.15)
Vaizey incontinence score	9.37	(4.90)	8.88	(5.26)	7.81	(4.50)	9.00	(4.95)	9.25	(4.88)	8.90	(5.19)
IBD fatigue score	8.66	(3.49)	8.67	(3.43)	9.29	(4.19)	9.85	(3.54)	8.71	(3.55)	8.93	(3.49)
IBD control score	8.81	(4.13)	8.84	(4.48)	9.68	(4.50)	8.59	(4.52)	8.88	(4.16)	8.79	(4.48)
IBD control VAS score	6.54	(2.13)	6.40	(2.21)	7.19	(1.68)	6.70	(2.09)	6.59	(2.10)	6.47	(2.19)
EQ5D utility score	0.73	(0.22)	0.73	(0.22)	0.76	(0.22)	0.71	(0.22)	0.73	(0.22)	0.73	(0.22)
Illness perceptions	42.46	(11.25)	42.62	(12.02)	41.35	(12.92)	43.93	(12.48)	42.37	(11.38)	42.91	(12.12)
Self-efficacy for managing chronic disease	5.91	(2.11)	5.82	(1.95)	5.82	(2.19)	5.36	(2.01)	5.90	(2.11)	5.72	(1.97)
All-or-nothing thinking	13.15	(5.08)	13.42	(4.76)	13.87	(4.88)	13.94	(5.12)	13.21	(5.06)	13.53	(4.84)
Avoidance/resting behaviour	18.52	(5.67)	19.01	(5.99)	16.94	(4.64)	17.97	(5.48)	18.39	(5.61)	18.78	(5.89)
Patient Health Questionnaire-9 (for depression)	9.36	(5.74)	9.10	(5.74)	8.16	(4.89)	10.09	(5.34)	9.26	(5.68)	9.32	(5.66)
Visceral Sensitivity Index	39.37	(16.35)	38.56	(17.75)	38.68	(16.77)	40.36	(15.28)	39.31	(16.37)	38.96	(17.24)
IBD status (Calprotectin & IBD control score)												
Remission	45	15.20%	61	24.11%	4	44.44%	9	21.95%	49	16.07%	70	23.81%
Active	251	84.80%	192	75.89%	5	55.56%	32	78.05%	256	83.93%	224	76.19%
IBD status (Calprotectin only)												
Remission	234	79.05%	211	83.40%	8	88.89%	32	78.05%	242	79.34%	243	82.65%
Active	62	20.95%	42	16.60%	1	11.11%	9	21.95%	63	20.66%	51	17.35%
Depression status												
Not depressed	200	55.87%	176	57.70%	21	67.74%	43	50.00%	221	56.81%	219	56.01%
Depressed	158	44.13%	129	42.30%	10	32.26%	43	50.00%	168	43.19%	172	43.99%
Rome IV criteria												
No	191	53.35%	162	53.29%	16	51.61%	38	44.19%	207	53.21%	200	51.28%
Yes	167	46.65%	142	46.71%	15	48.39%	48	55.81%	182	46.79%	190	48.72%

Mobility (EQ5D)												
No problems	222	63.61%	207	69.23%	2	100.00%	0	.%	224	63.82%	207	69.23%
Slight problems	59	16.91%	54	18.06%	0	0.00%	0	.%	59	16.81%	54	18.06%
Moderate problems	48	13.75%	24	8.03%	0	0.00%	0	.%	48	13.68%	24	8.03%
Severe problems	18	5.16%	14	4.68%	0	0.00%	0	.%	18	5.13%	14	4.68%
Unable	2	0.57%	0	0.00%	0	0.00%	0	.%	2	0.57%	0	0.00%
Self-care (EQ5D)												
No problems	280	80.23%	260	87.25%	2	100.00%	0	.%	282	80.34%	260	87.25%
Slight problems	35	10.03%	21	7.05%	0	0.00%	0	.%	35	9.97%	21	7.05%
Moderate problems	25	7.16%	16	5.37%	0	0.00%	0	.%	25	7.12%	16	5.37%
Severe problems	8	2.29%	1	0.34%	0	0.00%	0	.%	8	2.28%	1	0.34%
Unable	1	0.29%	0	0.00%	0	0.00%	0	.%	1	0.28%	0	0.00%
Usual activities (EQ5D)												
No problems	166	47.56%	167	56.04%	2	100.00%	0	.%	168	47.86%	167	56.04%
Slight problems	103	29.51%	72	24.16%	0	0.00%	0	.%	103	29.34%	72	24.16%
Moderate problems	53	15.19%	37	12.42%	0	0.00%	0	.%	53	15.10%	37	12.42%
Severe problems	23	6.59%	17	5.70%	0	0.00%	0	.%	23	6.55%	17	5.70%
Unable	4	1.15%	5	1.68%	0	0.00%	0	.%	4	1.14%	5	1.68%
Pain/discomfort (EQ5D)												
None	97	27.79%	91	30.54%	0	0.00%	0	.%	97	27.64%	91	30.54%
Slight	148	42.41%	135	45.30%	1	50.00%	0	.%	149	42.45%	135	45.30%
Moderate	86	24.64%	55	18.46%	1	50.00%	0	.%	87	24.79%	55	18.46%
Severe	13	3.72%	13	4.36%	0	0.00%	0	.%	13	3.70%	13	4.36%
Extreme	5	1.43%	4	1.34%	0	0.00%	0	.%	5	1.42%	4	1.34%
Anxiety/depression (EQ5D)												
None	140	40.11%	149	50.00%	1	50.00%	0	.%	141	40.17%	149	50.00%
Slight	123	35.24%	89	29.87%	1	50.00%	0	.%	124	35.33%	89	29.87%
Moderate	67	19.20%	45	15.10%	0	0.00%	0	.%	67	19.09%	45	15.10%
Severe	10	2.87%	12	4.03%	0	0.00%	0	.%	10	2.85%	12	4.03%
Extreme	9	2.58%	3	1.01%	0	0.00%	0	.%	9	2.56%	3	1.01%

Mean & standard deviation (SD) for continuous variables indicated by parentheses around SD; Absolute frequencies & column-wise percentages for categorical data indicated by '%' for percentage summary data

*in the last seven days